

The Foundation of Projection Relativity IV

The Global Singleton Theorem Axiom-by-Axiom Uniqueness and Canonical Closure of the Complete Projection Relativity Architecture

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Abstract

Projection Relativity manuscripts I through III present a no-fit spectral-projection architecture spanning radial, compact, electroweak, flavor, radiative, neutrino, and strong sectors. This paper proves equation-architecture uniqueness over explicitly declared admissible classes, modulo gauge, basis, coordinate, sign, algebraic, and unit equivalences. Direct ancestry from the seven PR-I postulates and proved canonical logical closure are the two accepted provenance routes. A sector may close through a canonical logical extension when compatibility, minimality, covariance, target-independent normalization, and exhaustive classification select one equivalence class without a fitted coefficient or new adjustable scale. A normal-form exhaustion theorem makes the equation-level claim precise: once the structural candidate class is fixed independently, every in-class equation either has a nonzero exact closure defect or is related to the canonical equation by an invertible ledger-preserving transformation. Merely matching one reported number is not counted as equivalence.

Exact rank and semigroup arguments establish rigidity of the minimal even quartic radial branch and parts of the compact recurrence. Formal coefficient extraction and rank-nullity fix $\sigma_8 = -1$ and $N_{\text{gen}} = 3$ inside explicitly declared boundary and sheet-realization classes. For neutral D_3 , exact rank reduction leaves $D_3(k) = (\epsilon_V^2/N_{\text{gen}})kP_Z$. The remaining response is fixed by a canonical-closure theorem. The unitary-basis invariance, orthogonal-mode additivity, quadratic homogeneity, and unit-charge normalization uniquely force the response functional to be $F(q) = \text{Tr}(q^\dagger q)$. Since the normalized charged-adjoint operator has eigencharges $(+1, -1)$, $F(q) = 2$ and the sole member of the declared minimal PR neutral-response class is $D_3 = 2(\epsilon_V^2/N_{\text{gen}})P_Z$. No empirical target or free coefficient enters this selection.

Abstract second-jet families and the gauge-invariant operator $|\Phi^\dagger D_\mu \Phi|^2$ lie outside the PR admissible class because they introduce independent response data or a new Wilson coefficient. They are therefore different theories, not additional PR parameters or alternative PR solutions. The complete PR class used here is fixed by the public PR-I through PR-III ledger together with inheritance, minimality, covariance, completeness, nonredundancy, canonical normalization, declared finite-tier order, and the no-fit firewall. On every equation-selection coordinate of that class, exact normal-form defects have one zero. We show that the dependency-ordered admissible equation quotient is the singleton $\{[\text{PR}]\}$: every working in-class presentation is equivalent, and every inequivalent proposal violates at least one declared guideline. No discrete or continuous free selector remains. This is a closure theorem for the PR equation architecture; distinct physical states of one fixed field equation are not distinct theories and are not counted as alternative equation architectures.

1 Introduction

PR-I defines the seven-postulate spectral-projection foundation [3]; PR-II supplies the non-Abelian electroweak and flavor lift [4]; and PR-III supplies a finite-tier radiative and cross-sector closure with an explicit no-fit provenance firewall [5]. PR-IV closes the equation-equivalence question for the declared PR admissible class. The public PR-I through PR-III equations and finite-tier rules form the inherited ledger; the present theorem proves that their admissible equation-selection quotient contains one physical equivalence class.

The absolute rule is the no-fit firewall. No diagnostic residual may generate an equation, coefficient, branch, or admissibility condition. Direct derivation from the PR-I postulates is one complete provenance route, and a downstream relation may also be admitted as a canonical logical closure when its framework origin is stated, its candidate class is declared, and exact classification leaves no adjustable freedom. An inequivalent construction enters the uniqueness quotient only when it satisfies that same admissible class. A construction that introduces an independent operator, response jet, or coefficient belongs to an enlarged theory and maps the boundary of the PR closure class without modifying the canonical PR branch. There is no unresolved selector inside the class defined in this paper.

All public-source statements are audited against repository commit 9680f70b21f0 (full hash in the source ledger). Machine-readable exact certificates accompany the manuscript [2].

2 Foundational Derivation and Canonical Logical Closure

The statement that the PR equations are the only physical solution must be defined over a mathematical candidate class. For any fixed number r , infinitely many syntactically different equations can be constructed with r as a root, for example

$$x - r = 0, \quad (x - r)(x^2 + 1) = 0, \quad e^{x-r} - 1 = 0. \quad (1)$$

Invertible field redefinitions generate further equivalent descriptions. Consequently, unrestricted uniqueness among every imaginable written equation is not a meaningful theorem.

The three displayed equations do not constitute three physical solutions over the real line. The second is the first multiplied by the nowhere-zero factor $x^2 + 1$, while the third obeys

$$e^{x-r} - 1 = (x - r) \int_0^1 e^{t(x-r)} dt, \quad (2)$$

and the integral is strictly positive. Thus all three are invertible re-presentations of the same residual equation. This example is the model for the equivalence proof below: different typography is not a new equation when a nowhere-singular transformation preserves the complete operator and solution content.

Let $\mathcal{P}_0$ be the PR-I foundational postulates and let $\mathcal{C}_{\text{PR}}$ be the set of explicitly declared canonical closure rules used to extend that foundation into a downstream sector. Let $\mathfrak{A}_{\text{PR}}(\mathcal{P}_0, \mathcal{C}_{\text{PR}})$ denote the complete class of operators satisfying the inherited topology, regularity, symmetry, stability, representation, boundary, and finite-tier conditions. Let $\sim$ identify gauge, basis, coordinate, phase, algebraic, and unit-equivalent descriptions. The meaningful physical uniqueness target is

$$\boxed{\text{Sol}(\mathfrak{A}_{\text{PR}}(\mathcal{P}_0, \mathcal{C}_{\text{PR}})) / \sim = \{\{\text{PR}\}\}}. \quad (3)$$

Direct derivation from $\mathcal{P}_0$ and proved canonical logical closure are the two complete admissible provenance routes used in this paper. Requiring every downstream operator to appear literally in the initial short postulate list would confuse a foundation with its uniquely constrained mathematical development.

Every downstream object in the closed theorem domain receives one of five final mathematical statuses:

1. a *direct postulate theorem*, proved from $\mathcal{P}_0$;
2. a *derived definition*, which conservatively abbreviates existing objects and adds no models;
3. a *canonical logical closure*, uniquely selected from inherited PR structures by explicit consistency and minimality conditions; or
4. an *equivalent presentation*, related to the canonical equation by an allowed invertible ledger-preserving transformation; or
5. an *outside-class construction*, which violates at least one defining admissibility condition and is therefore a different theory rather than a PR solution.

A fitted ansatz is not an acceptable status: empirical diagnostics are forbidden from generative ancestry.

Proposition 2.1 (Canonical-closure admissibility standard). *A downstream rule C is an admissible canonical PR closure when all of the following are proved or explicitly verified:*

1. **inheritance:** C preserves the fields, representations, symmetries, normalizations, and already closed sector relations on which it acts;
2. **necessity:** C fills a specifically identified operator, projection, boundary, or representation map required to continue the PR construction;
3. **minimality:** C introduces no unnecessary field, independent scale, Wilson coefficient, response datum, or branch label;
4. **covariance:** its statement is invariant under every quotient equivalence appropriate to the sector;
5. **class-relative uniqueness:** exhaustive analytic or certified classification leaves one equivalence class in the declared candidate set; and
6. **target independence:** no measured value, residual, or diagnostic tolerance selects its form or normalization.

Such a closure is a logically controlled extension of the framework, not an additional adjustable physical postulate.

Proposition 2.2 (Complete certified PR admissible class). *Let $\mathfrak{A}_{\text{PR}}^{\text{can}}$ be the dependency-ordered finite-tier class obtained by fixing the public PR-I–PR-III field, representation, projection, boundary, and operator ledger and imposing the following closure guidelines on every equation-selection coordinate:*

1. inheritance of every previously fixed identity and normalization;
2. the declared PR field and representation inventory, with no new independent field, operator coefficient, scale, response datum, or branch label;
3. covariance under the complete gauge, normalized-basis, coordinate, phase, algebraic, and unit quotient;
4. the declared finite-tier order and framework-native projection grammar;
5. minimality, completeness, and nonredundancy at that order;
6. canonical unit normalization of each retained mode or response; and
7. target independence under the no-fit firewall.

The equation-selection coordinates of this class are the minimal radial coefficients, minimal required-channel projector, signed terminal boundary, nonzero-ledger hypercharge orbit, photon null line of the fixed neutral mass operator, normalized generation incidence, and fixed-kernel normalized neutral response. All other public ledger maps are frozen inherited relations rather than adjustable selector coordinates. Thus $\mathfrak{A}_{\text{PR}}^{\text{can}}$ is the complete class to which the uniqueness theorem in this paper applies; an operator obtained by relaxing any listed guideline is not a second PR solution.

Freezing an inherited sector map is not a hidden fit or a free parameter: it identifies which theory is being proved unique. The uniqueness question is whether any inequivalent equation choice remains after that public ledger and the independently stated closure guidelines are imposed. None does. Changing a frozen map, adding an operator, or changing the finite-tier grammar defines a different theory and cannot be counted as a second solution of the same PR system. Accordingly, the companion provenance graph permits either of the paths

$$\mathcal{P}_0 \longrightarrow E_1 \longrightarrow \cdots \longrightarrow R, \quad \text{or} \quad \mathcal{P}_0 \longrightarrow E_1 \longrightarrow C_{\text{can}} \longrightarrow \cdots \longrightarrow R, \quad (4)$$

provided a canonical-closure node carries the proof obligations in Proposition 2.1. The graph must remain acyclic, every parent must exist, and no empirical diagnostic may be an ancestor of a generated node. Postulate-only reachability and proved canonical-closure reachability are reported as complementary provenance levels. The PR-I postulate list remains protected against arbitrary promotion of a successful formula. The no-fit rule gives the final admissibility dichotomy:

$$\boxed{\begin{array}{l} \mathcal{P}_0 + \text{closure criteria} \longrightarrow C_{\text{can}} \text{ is admissible,} \\ \text{independent coefficient, operator, scale, or response datum} \longrightarrow \text{outside } \mathfrak{A}_{\text{PR}}^{\text{can}}. \end{array}} \quad (5)$$

This distinction preserves the explanatory force of the foundation without imposing the unreasonable requirement that the initial postulates explicitly encode every later sector equation.

3 No-Other-Solutions Standard

For each sector s , let $\mathfrak{A}_s$ be the explicitly declared admissible candidate set, let $\sim_s$ identify gauge, basis, phase, coordinate, algebraic, and unit conventions, and let F_s collect the target-independent mathematical and closure constraints. The meaningful uniqueness claim requires

$$\boxed{\{x \in \mathfrak{A}_s : F_s(x) = 0\} / \sim_s = \{[x_{\text{PR}}]\}}. \quad (6)$$

Verifying only $F_s(x_{\text{PR}}) = 0$ establishes existence, not uniqueness. The proof must also show local isolation after unphysical null directions are removed and exhaust every disconnected branch *inside the same declared class*. A model in a broader theory belongs to a different theorem domain unless it also satisfies all of F_s . Conversely, if two inequivalent members satisfy every declared condition, the admissible quotient contains two classes rather than the singleton required by Eq. (6).

3.1 Equation Equivalence and Exhaustive Uniqueness

Fix a sector s . Let $\mathcal{X}_s$ be its configuration space after all upstream fields, domains, normalizations, and identities have been fixed; let $\mathcal{Y}_s$ be its residual space; and let $\mathcal{G}_s$ be the declared group of gauge transformations, normalized basis changes, coordinate and unit changes, and invertible field redefinitions that preserve the frozen ledger and registered observable algebra. A candidate equation is a covariant residual map

$$E : \mathcal{X}_s \longrightarrow \mathcal{Y}_s, \quad Z(E) := \{x \in \mathcal{X}_s : E(x) = 0\}. \quad (7)$$

The structural candidate class $\mathfrak{E}_s^0$ is fixed before its equations are solved and is saturated under $\sim_{\text{op}}$, so an invertible rewriting cannot be excluded merely by typography. It is specified by field content, operator domain, locality or finite-tier order, regularity, covariance, boundary grammar, representation content, and inherited normalizations. A target-independent closure ledger is a map

$$\Delta_s : \mathfrak{E}_s^0 \longrightarrow [0, \infty)^{m_s}. \quad (8)$$

An equation is admissible precisely when $\Delta_s(E) = 0$. Neither $\mathfrak{E}_s^0$ nor Δ_s may contain a measured target, diagnostic residual, fitted tolerance, or the desired answer inserted as a coefficient. Unit normalization, zero incidence, completeness, nonredundancy, covariance, and fixed-order minimality are admissible only when stated independently of the value obtained after the equations are solved. Three relations must be distinguished:

1. $E \sim_{\text{op}} E'$, called *operator equivalence*, when there are $g \in \mathcal{G}_s$ and $A_x \in \text{GL}(\mathcal{Y}_s)$ at every x such that

$$E'(gx) = A_x E(x) \quad \text{for every } x \in \mathcal{X}_s. \quad (9)$$

This includes algebraic rearrangement, multiplication by a nowhere-zero factor, and allowed invertible changes of variables.

2. $E \sim_{\text{sol}} E'$, called *solution equivalence*, when an allowed transformation maps the complete zero set of one equation bijectively onto that of the other, $gZ(E) = Z(E')$.
3. $E \sim_{\text{phys}} E'$, called *physical equivalence*, when their quotient solution spaces are isomorphic and the isomorphism preserves every registered observable, multiplicity, boundary label, and physical spectral datum:

$$Z(E)/\mathcal{G}_s \cong Z(E')/\mathcal{G}_s. \quad (10)$$

Operator equivalence implies solution equivalence, and solution equivalence under an allowed ledger-preserving transformation implies physical equivalence. The converses need not hold. In particular, matching one reported number is not equivalence. Equations must agree on the complete admissible domain and preserve the physical operator, not merely one diagnostic evaluation. For example, $u = 0$ and $u^3 = 0$ have the same real zero set, but their linearizations at the zero have different rank and they are not operator-equivalent through an invertible residual multiplier.

Theorem 3.1 (Normal-form exhaustion and alternative-equation trichotomy). *Suppose a sector has a parameter space Θ_s and a normal-form map $\nu_s : \mathfrak{E}_s^0 \rightarrow \Theta_s$ with the following independently proved properties:*

1. ν_s is complete: $\nu_s(E) = \nu_s(E')$ if and only if $E \sim_{\text{op}} E'$;
2. admissibility is decided in normal form by a target-independent certificate $d_s : \Theta_s \rightarrow [0, \infty)^{m_s}$,

$$\Delta_s(E) = 0 \iff d_s(\nu_s(E)) = 0; \quad (11)$$

3. the certificate has exactly one zero, $d_s^{-1}(0) = \{\theta_s^*\}$.

Then

$$\boxed{\{E \in \mathfrak{E}_s^0 : \Delta_s(E) = 0\} / \sim_{\text{op}} = \{[E_s^*]_{\text{op}}\}}, \quad (12)$$

and every admissible equation induces the same physical quotient solution space as E_s^* . If, in addition, the canonical equation has an isolated physical solution,

$$Z(E_s^*)/\mathcal{G}_s = \{[x_s^*]\}, \quad (13)$$

then the sector has one physical solution class as well. Every proposed equation $\tilde{E}$ has exactly one of three physical dispositions: it is physically equivalent to E_s^* on the complete declared domain and is not a second physical equation; it is physically inequivalent but lies in $\mathfrak{E}_s^0$, in which case it has a nonzero defect; or it is physically inequivalent and lies outside $\mathfrak{E}_s^0$, in which case it belongs to an enlarged theory outside the theorem. An inequivalent in-class equation satisfying every declared condition enlarges the admissible quotient and must therefore be included in the theorem statement.

Proof. Let E be admissible. The certificate property gives $d_s(\nu_s(E)) = 0$, so the unique-zero property gives $\nu_s(E) = \theta_s^*$. If E_s^* is any representative with that normal form, completeness gives $E \sim_{\text{op}} E_s^*$. Hence all admissible equations occupy one operator-equivalence class. Equation (9) maps their zero sets bijectively and uses only transformations in $\mathcal{G}_s$, so their physical quotient solution spaces agree. If Eq. (13) also holds, each of those spaces is the same singleton. The three dispositions are the same argument and its contrapositive for in-class equations. For an outside proposal, a full-domain physical-equivalence test places it in the first disposition if it is merely another representation; otherwise its inequivalent operator content places it in the enlarged-theory disposition. $\square$

The theorem is noncircular: the candidate class, quotient group, normal-form classification, and defect ledger are fixed without using θ_s^* . That value first appears as the computed zero of the certificate. The substantive proof obligation is therefore to classify every in-class equation into a faithful normal form and then exhaust the complete zero set of its defect.

Finite discrete candidates are treated by exhaustive enumeration. Polynomial and rational constraints are solved by exact elimination or complete analytic parameterization. Continuous numerical roots require interval isolation and exclusion of the complement. Operator claims require self-adjoint domains and certified spectral enclosures. Radiative corrections require a complete invariant-operator basis and an explicit power-counting rule.

Post-generation diagnostic agreement is not a generative admissibility constraint. If $\mathcal{E}_s$ denotes external data and $\mathcal{C}_s$ the construction map, the firewall condition is

$$\mathcal{E}_s \cap \text{Dom}(\mathcal{C}_s) = \emptyset. \quad (14)$$

No residual or standard-deviation score may select a branch, coefficient, truncation order, numerical resolution, or acceptance tolerance.

Thus “no other solution” never means that no differently written equation can return the same number. It means that no inequivalent operator or branch in the declared PR closure class satisfies the complete structural ledger. The global theorem may combine direct postulate theorems and proved canonical closures, provided their status is explicit and every included sector contains one zero-defect equivalence class.

4 Rigidity of the PR-I Radial and Compact-Boundary Block

The first uniqueness problem is upstream of every electromagnetic, electroweak, flavor, and radiative result. It asks whether the PR-I radial operator and compact boundary map are fixed by the declared projection rules. This section proves rigidity in the completed PR-I admissible class and assigns a nonzero admissibility defect to every inequivalent in-class value.

4.1 Minimal Radial Candidate Class

Let

$$\mathfrak{V}_4 = \left\{ V(w) = \Lambda_X^2 \left(1 + a_2 w^2 + a_4 w^4 \right) \mid a_2 > 0, a_4 > 0 \right\} \quad (15)$$

be the minimal even quartic class. Unit local radial stiffness gives

$$\frac{1}{2\Lambda_X^2} \frac{d^2 V}{dw^2} \Big|_{w=0} = a_2 = 1, \quad (16)$$

whereas the spatial projection trace gives

$$a_4 = \frac{\text{Tr } P_{\text{space}}}{\text{Tr } I_{3+1}} = \frac{3}{4}. \quad (17)$$

The two coefficient constraints are independent; their coefficient matrix has determinant $4 \neq 0$. Consequently, there is exactly one member of $\mathfrak{V}_4$ satisfying both constraints:

$$\boxed{V_{X,\star}(w) = \Lambda_X^2 \left(1 + w^2 + \frac{3}{4} w^4 \right)}. \quad (18)$$

Furthermore,

$$\frac{V''_{X,\star}(w)}{\Lambda_X^2} = 2 + 9w^2 > 0, \quad \lim_{|w| \rightarrow \infty} V_{X,\star}(w) = +\infty. \quad (19)$$

Thus the associated one-dimensional Schrödinger operator is confining and has a lower-bounded discrete simple spectrum on its standard self-adjoint domain [7].

Equation (18) is unique in the class $\mathfrak{V}_4$. Higher even or nonpolynomial terms violate the declared minimal-even-quartic rule and define a different operator class; they are not additional PR radial solutions.

4.2 Interior Recurrence as a Numerical Semigroup

The primitive compact returns have lengths 3 and 4. Every recurrent interior length belongs to

$$S_{3,4} = \langle 3, 4 \rangle = \{3a + 4b \mid a, b \in \mathbb{Z}_{\geq 0}\}. \quad (20)$$

The positive gaps of this semigroup are exactly

$$\mathbb{Z}_{>0} \setminus S_{3,4} = \{1, 2, 5\}. \quad (21)$$

Indeed, $6 = 3 + 3$, $7 = 3 + 4$, and $8 = 4 + 4$ cover all residue classes modulo 3; adding copies of 3 proves that every integer greater than or equal to 6 lies in $S_{3,4}$. A finite-rank boundary word must contain the leave–return pair A^2 followed by the primitive spatial return A^3 . It therefore has exponent at least 5. Equation (21) then proves that

$$XA^5 = XA^{2+3} \quad (22)$$

is the unique positive boundary word that closes and is not already generated by the recurrent interior monoid.

The first repeated full-trace return is $A^{4+4} = A^8$. Under the PR finite-rank orientation rule retain the spatial primitive channel and subtract the first double full-trace recirculation outside the finite slice the signed exclusion is $-XA^8$. This exponent and sign are unique under that orientation rule. The semigroup alone does not derive the rule; the rule remains an explicit geometric condition of the present theorem. The resulting signed terminal set is

$$\boxed{\text{Adm}_{\Pi_{13}} = \{\epsilon, A^3, A^4, XA^5, -XA^8\}}. \quad (23)$$

4.3 Exact Cofactor Closure

With $T_A = 1/4$ and $T_X = 3/4$,

$$M_A = \begin{pmatrix} T_A^3 & T_A^4 \\ 1 & 0 \end{pmatrix} \quad (24)$$

gives

$$D_A = \det(I - M_A) = 1 - T_A^3 - T_A^4 = \frac{251}{256}. \quad (25)$$

The signed terminal set gives

$$N_{bc} = 1 + T_A^3 + T_A^4 + T_X(T_A^5 - T_A^8) = \frac{267453}{262144}. \quad (26)$$

Consequently,

$$c_{bc} = T_X \left[1 + T_A^2 - T_A^6 \frac{N_{bc}}{D_A} \right] = \frac{3354902985}{4211081216} = 0.7966844648478990532 \dots \quad (27)$$

These are rational identities and contain no empirical input.

4.4 Minimal Finite-rank Projector

Let the odd radial eigenbasis be $\{\phi_1, \phi_3, \phi_5, \dots\}$. Require a coordinate spectral projector that contains both the fundamental odd winding-linked channel ϕ_1 and the spatial primitive-return channel ϕ_3 , preserves parity, and has minimal rank. There is exactly one rank-two coordinate projector satisfying these requirements:

$$\Pi_{13} = |\phi_1\rangle\langle\phi_1| + |\phi_3\rangle\langle\phi_3|. \quad (28)$$

The theorem domain requires channels (1) and (3), as fixed by their fundamental winding-linked and primitive spatial-return roles. The minimal-channel principle is the canonical PR-I closure: every added odd channel violates minimal rank and lies outside the admissible class.

4.5 Two-mode Spectral Isolation and Leakage

Define

$$K_X = \frac{O_X - \lambda_0}{\lambda_1 - \lambda_0} \quad (29)$$

and

$$P_{\text{geom}}^{\text{bc}} = \Pi_{13} K_X^2 \left(c_{bc} + w^2 + \frac{3}{4} w^4 \right) K_X^2 \Pi_{13}. \quad (30)$$

On the range of Π_{13} , Eq. (30) is a real symmetric 2×2 matrix. Its reconstructed off-diagonal element is nonzero and its eigenvalues are distinct. The dominant normalized eigenvector is therefore unique up to an overall sign. The sign cancels from

$$p_1 = |\langle\phi_1|\Omega_{bc}\rangle|^2, \quad p_3 = 1 - p_1, \quad (31)$$

so the leakage probabilities are unique in the two-mode block. The independent harmonic-oscillator-basis certificate gives

$$\lambda_0 = 2.322872581463 \dots, \quad (32)$$

$$\lambda_1 = 5.375889248196 \dots, \quad (33)$$

$$\lambda_3 = 13.233893408730 \dots, \quad (34)$$

$$\mu_{\text{min}}^2 = 3.053016666733 \dots, \quad (35)$$

$$p_1 = 7.68534695089 \times 10^{-4}. \quad (36)$$

No empirical electromagnetic constant enters this calculation. The compact stiffness and tree normalization are generated as

$$q_{bc} = p_1(\lambda_1 - \lambda_0) + (1 - p_1)(\lambda_3 - \lambda_0) = 10.904981678435\dots, \quad (37)$$

$$\alpha_{PR,tree}^{-1} = 4\pi q_{bc} = 137.036041314016\dots \quad (38)$$

This calculation reproduces the frozen PR-III high-resolution branch. It does not reproduce the earlier finite-resolution PR-I value $p_1 = 7.6528903366 \times 10^{-4}$. The latter is therefore classified as superseded finite-resolution data rather than as the canonical operator limit.

4.6 PR-I Closure Theorem

The radial coefficient solution, numerical-semigroup gap theorem, positive terminal word, rational cofactor arithmetic, and minimal coordinate projector are exact in the declared class. The two-mode leakage and high-resolution compact stiffness are numerically isolated and reproducible. The minimal-quartic function class, primitive return set $\{A^3, A^4\}$, required channel identification $\{\phi_1, \phi_3\}$, and finite-rank orientation rule producing $-XA^8$ are the declared minimal closure conditions. They define the complete PR-I theorem domain used in this paper. Within that domain, the radial and compact-boundary equation quotient is a singleton.

5 Compact-sector Independence and Minimal Closure Requirements

The seven PR-I postulates establish the common geometric foundation. The compact construction is completed by the PR-I boundary-orientation and spectral-minimality principles. This section formalizes those sector-specific conditions and proves that they select the released compact equations without a diagnostic input. We hold fixed the master field, internal cylinder, metric, Hilbert space, separable spectral operator, projection-trace radial potential, trace alphabet $T_A = 1/4$, $T_X = 3/4$, and the primitive-return denominator

$$D_A = 1 - T_A^3 - T_A^4 = \frac{251}{256}. \quad (39)$$

No diagnostic input is admitted.

For $\sigma \in \{-1, 0, +1\}$ define the finite, integer-coefficient terminal cofactor

$$N_\sigma = 1 + T_A^3 + T_A^4 + T_X (T_A^5 + \sigma T_A^8). \quad (40)$$

Each member contains no continuous free parameter. Nevertheless the three members yield distinct rational values of N_σ , $R_\sigma = N_\sigma/D_A$, and

$$c_\sigma = T_X (1 + T_A^2 - T_A^6 R_\sigma). \quad (41)$$

The canonical PR coefficient is $\sigma = -1$. The pre-projection exclusion equation has exact defect $|1 + \sigma|$, so $\sigma = -1$ is its sole zero; the other displayed coefficients are comparison values outside the completed PR-I boundary class. For exhaustive comparison in the nondegenerate radial eigenbasis, consider

$$\Pi_{1j} = |\phi_1\rangle\langle\phi_1| + |\phi_j\rangle\langle\phi_j|, \quad j \in \{3, 5, 7\}, \quad (42)$$

Each is a rank-two, parity-preserving orthogonal spectral projector with no continuous coefficient. The projector comparison is resolved by the required-channel minimality rule. Among coordinate projectors containing the fundamental winding-linked channel ϕ_1 and primitive spatial-return channel ϕ_3 , minimal rank uniquely gives Π_{13} . Projectors containing different odd channels do not satisfy that declared PR-I closure condition.

Thus the signed-incidence and required-channel minimality certificates each have one admissible PR orbit. Their Cartesian certificate classifies all nine displayed coefficient/projector combinations and retains only $(-1, \Pi_{13})$. These target-independent closure principles are logical extensions of the PR foundation and need not be literal restatements of Postulates 1 through 7.

6 PR-II Exact Classifications and Canonical Closure

PR-II extends the PR-I foundation to an electroweak and flavor architecture [4]. The direct-ancestry audit separates exact consequences of a fixed electroweak representation from the logical closure principles used to construct that representation.

6.1 Hypercharge Solution within Fixed Matter Content

Write the one-generation left-handed Weyl hypercharges [8] as q, u, d, ℓ, e for the quark doublet, up conjugate, down conjugate, lepton doublet, and charged-lepton conjugate. With $q = t$, the linear local-anomaly equations reduce to

$$\ell = -3t, \quad e = 6t, \quad d = -2t - u. \quad (43)$$

Substitution into the cubic anomaly gives the exact factorization

$$-18t(u - 2t)(u + 4t) = 0. \quad (44)$$

For $t \neq 0$, the last two factors yield the standard branch and its up/down singlet relabeling. Overall hypercharge normalization is conventional. The $t = 0$ factor gives an anomaly-free vectorlike singlet family $q = \ell = e = 0$, $d = -u$; it is excluded only when the nonzero residual-charge ledger is imposed. Hence anomaly cancellation alone is not a unique derivation, whereas anomaly cancellation plus a fixed nonzero charge ledger does isolate the familiar branch modulo normalization and labeling.

6.2 Unique Photon Direction in the Declared Order Representation

Once the neutral mass matrix is derived in outer-product form,

$$M_0^2 \propto \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix} = \begin{pmatrix} g \\ -g' \end{pmatrix} \begin{pmatrix} g & -g' \end{pmatrix}, \quad (45)$$

nonzero g, g' imply rank one and a one-dimensional nullspace. The photon direction is unique up to normalization. The outer-product form is a fixed inherited PR-II representation relation. Other symmetric rank-one matrices violate that representation ledger and are outside the declared class.

6.3 Generation Replication Under Anomaly Gates Alone

Let one complete anomaly-free generation contain four weak doublets after color multiplicity. Replicating the complete ledger N times multiplies every local anomaly coefficient by N , so all remain zero. The number of weak doublets is $4N$, which is even for every $N \in \mathbb{Z}_{>0}$; the Witten [10] parity condition therefore also holds. Consequently,

$$N = 1, 2, 3, \dots \quad (46)$$

is an infinite discrete family satisfying these gates. Anomaly and Witten conditions alone therefore do not select $N_{\text{gen}} = 3$. Appendix E supplies the additional PR-II completeness/nonredundancy sheet-incidence theorem, exhausts every positive integer, and uniquely yields $N = 3$ in the declared realization. The normalized-incidence condition $\Phi_3^\dagger \Phi_3 = I_3$ then leaves one unitary operator orbit, as proved in Appendix D.

Its logical justification is minimal faithful realization: every broken compact direction must be represented (surjectivity), while no invisible sheet combination may remain (injectivity). These are target-independent closure conditions for the PR-II lift; literal appearance in P1 through P7 is not required.

The electroweak lift, weak boundary-deficit rule, hierarchy action, flavor overlaps, neutrino branch, and strong anchor likewise admit either direct ancestry proofs or explicit target-independent closure principles. The graph records those two routes as complementary provenance classifications for the PR-II extension.

7 PR-III Finite-tier Uniqueness and Closure Classification

PR-III reports a reproducible no-fit radiative and cross-sector closure at its declared finite tier [5]. That order is a defining condition of the PR class rather than a parameter varied inside it. PR-IV classifies equation presentations at that fixed order.

7.1 Electromagnetic Radiative Monomials

The PR-III electromagnetic corrections use dimensionless factors drawn from p_1 , Δ_{EW} , s_W^2 , and $N_{\text{gen}}^{-1/2}$, including

$$D_1 = -p_1 \Delta_{\text{EW}} s_W^2, \quad (47)$$

$$D_2 = D_1 \Delta_{\text{EW}} N_{\text{gen}}^{-1/2}. \quad (48)$$

If one drops the declared PR-III finite-tier composition grammar, one can write formal expressions such as

$$-p_1 \Delta_{\text{EW}} s_W^2 N_{\text{gen}}^{-1/2}, \quad -p_1 \Delta_{\text{EW}}^2 s_W^2, \quad -p_1 \Delta_{\text{EW}} (s_W^2)^2 \quad (49)$$

from the same symbols. The bounded-degree audit enumerated 23 such expressions through total degree six. Twenty-one violate the inherited finite-tier composition order and therefore lie outside $\mathfrak{A}_{\text{PR}}^{\text{can}}$; they are not alternative PR equations. Inside the declared grammar, D_1 and D_2 are the sole two normal forms and carry no adjustable exponent or coefficient.

7.2 Charged and Neutral Electroweak Coefficients

The C_2 correction uses the charged/neutral coefficient pair $(2, -1)$ and the neutral D_3 correction uses coefficient 2. Before any representation-trace theorem is imposed, the preregistered sign gate alone is satisfied, for example, by $(1, -1)$, $(2, -1)$, and $(3, -2)$. Likewise the positive neutral coefficients 1, 2, 3, 4 all satisfy a strict positivity-only gate. The small-integer audit found 16 opposite-sign C_2 pairs and four positive D_3 coefficients in its stated range. More strongly, the exact real-symmetric gauge-index coefficient classification at the fixed D_3 Lorentz/momentum kernel in Appendix G shows that the three linear structural gates leave $D_3(k) = \rho k P_Z$ for every real k within that matrix class. Adding the separate strict positivity-only gate restricts this to $k > 0$, still an uncountable half-line. Fixed rational members are discrete, contain no fitted continuous coefficient, and are not convention-equivalent once kinetic and field normalization are frozen.

The PR-III pair and coefficient are therefore fixed by a framework-native projector-trace and response closure, not by the sign gate or diagnostic residual. Appendix G now proves that the canonical charged-adjoint contraction has neutral representation trace two and that the four canonical defects vanish only for $D_3 = 2\rho P_Z$. Theorem G.5 proves that the normalized charged-adjoint functional is the unique minimal response and therefore fixes $k = 2$ within the declared PR class. A determinant-Hessian expression satisfying the same normalized physical- Z map is an equivalent microscopic representation of this result. The all-real family obtained after omitting canonical normalization lies outside the PR class. The

locked M_Z gate permits an interval containing $k = 1, 2, 3$, confirming that the exact normalized-response defect, rather than residual ranking, supplies the selection. The C_1 transport rule, C_2 trace pair, and multiplicative neutrino envelope are fixed inherited finite-tier maps; changing their operator grammar defines a different theory rather than another solution of the declared PR equations.

The strong-sector one-loop coefficient is fixed once its representation and active-threshold inventory are imposed. Its theorem domain is exactly the declared one-loop strong class. Higher-order or nonperturbative operators lie outside that finite-tier class and are not additional solutions of it.

8 Completed Derivation Graph and Global Closure Theorem

The PR-I–PR-III construction is a dependency-ordered directed acyclic graph. Its public sector maps and finite-tier rules are inherited relations; its equation-selection coordinates are exactly those listed in Proposition 2.2. Every node in the claimed theorem domain is therefore primitive, derived, a proved canonical closure, a fixed inherited map, or an equivalent presentation. Diagnostic references have no generative edges. No unresolved selector is an element of the completed admissible class.

Table 1: Final Uniqueness Classification in the Completed PR Admissible Class.

Block	Final result
PR-I radial coefficients	Sole minimal even-quartic normal form $(a_2, a_4) = (1, 3/4)$.
PR-I compact projector	Sole minimal required-channel range $\text{span}\{\phi_1, \phi_3\}$.
PR-I terminal boundary	Sole signed zero-incidence value $\sigma_8 = -1$.
PR-II hypercharge	Sole nonzero-charge orbit modulo normalization and up/down relabeling.
PR-II photon	Sole null line of the fixed representation-derived outer-product neutral mass operator.
PR-II generations	Sole normalized complete/nonredundant incidence orbit $(N_{\text{gen}}, [\Phi_3]) = (3, [I_3])$.
PR-III finite-tier maps	D_1, D_2, C_1, C_2 and the declared neutrino/strong maps are fixed inherited finite-tier relations, not free selector coordinates.
PR-III neutral response	Sole normalized minimal-response tensor $D_3 = 2\rho P_Z$.
Provenance	No diagnostic target is a generative ancestor and no fitted coefficient occurs.
Complete admissible quotient	One dependency-ordered equation architecture modulo ledger-preserving equivalence.

Theorem 8.1 (Singleton Theorem for the Completed PR Admissible Class). *Let $\mathfrak{A}_{\text{PR}}^{\text{can}}$ be the complete class in Proposition 2.2, and let $\sim$ identify gauge, normalized-basis, coordinate, phase, algebraic, field-redefinition, and unit presentations that preserve the complete public ledger and registered observable algebra. Then*

$$\boxed{\mathfrak{A}_{\text{PR}}^{\text{can}} / \sim = \{[\text{PR}]\}}. \quad (50)$$

Equivalently, the joint exact closure-defect map has one zero orbit. Every proposed equation has exactly one final disposition: it is an equivalent presentation of the PR equation on the complete declared domain; it has a strictly positive in-class closure defect; or it violates a defining admissibility guideline and belongs to a different theory. Hence there is no second inequivalent PR equation architecture and no discrete or continuous free selector in the declared class.

Proof. The minimal even-quartic constraints have the sole coefficient pair $(a_2, a_4) = (1, 3/4)$. Minimal required-channel rank two fixes $\text{Ran } \Pi = \text{span}\{\phi_1, \phi_3\}$. Signed pre-projection incidence has the sole zero $|1 + \sigma_8| = 0$. The factored hypercharge ideal with the nonzero charge ledger leaves one orbit modulo

normalization and up/down relabeling, and the fixed rank-one neutral mass matrix has a one-dimensional photon kernel. Completeness, nonredundancy, and normalized incidence give $(N_{\text{gen}}, [\Phi_3]) = (3, [I_3])$. Finally, the normalized quadratic-response certificate $\rho^2(k-2)^2$ has the sole zero $k=2$.

These unique normal forms are fibers of the acyclic public dependency graph, not independent Cartesian choices. Every remaining PR-I–PR-III sector map is a fixed inherited relation of the class and therefore contributes no selector coordinate. Successive composition of fixed maps with singleton fibers remains a singleton. Operator-equivalent rewritings are removed by $\sim$; an inequivalent in-class proposal has a nonzero component of the joint defect; and a proposal with new operator data, scale, coefficient, order, or field inventory violates the class definition. These cases exhaust every proposal and prove Eq. (50). $\square$

The theorem concerns uniqueness of the PR equation architecture. Different states governed by one fixed field equation are solutions of the same unique theory, not alternative equation architectures. No claim over theories built from different postulates, fields, operator inventories, or power-counting rules is needed for the PR closure statement.

9 Conclusion: Closure of the PR Equation Architecture

PR-IV establishes the requested closing statement for the admissible PR-I through PR-III equation architecture. The theorem domain is fixed before any diagnostic comparison: the public finite-tier ledger is inherited, and every equation-selection coordinate must satisfy inheritance, covariance, minimality, completeness, nonredundancy, canonical normalization, and the no-fit firewall. Gauge, normalized-basis, coordinate, phase, algebraic, field-redefinition, and unit changes are quotiented as equivalent presentations.

The exact normal forms have sole admissible values. The radial coefficients are $(a_2, a_4) = (1, 3/4)$; the minimal required-channel projector retains $\text{span}\{\phi_1, \phi_3\}$; signed terminal incidence fixes $\sigma_8 = -1$; the nonzero hypercharge ledger leaves one orbit; the fixed neutral mass operator leaves one photon null line; normalized complete and nonredundant sheet incidence fixes $(N_{\text{gen}}, [\Phi_3]) = (3, [I_3])$; and the neutral response fixes $k=2$. For the last coordinate, the structural tensor is

$$D_3(k) = \rho k P_Z, \quad (51)$$

while unitary-basis invariance, orthogonal-mode additivity, quadratic homogeneity, and unit-charge normalization uniquely force

$$\mathcal{F}(q) = \text{Tr}(q^\dagger q). \quad (52)$$

The normalized charged-adjoint spectrum is $(+1, -1)$, so $\mathcal{F}(q) = 2$ and

$$\boxed{D_3 = 2\rho P_Z}. \quad (53)$$

This is not a best-fit choice: every $k \neq 2$ has the exact positive defect $\rho^2(k-2)^2$.

Combining the seven unique fibers in dependency order with the fixed inherited PR-I through PR-III maps gives

$$\boxed{\mathfrak{A}_{\text{PR}}^{\text{can}} / \sim = \{[\text{PR}]\}}. \quad (54)$$

There is therefore no second inequivalent equation, branch, normalization, or continuous parameter allowed by the complete PR admissibility guidelines. A differently written formula that works on the complete declared domain is a ledger-preserving presentation of the same physical equation. An inequivalent in-class proposal has a nonzero exact defect. A proposal that adds an operator, field, scale, Wilson [9] coefficient, response jet, higher order, or different boundary/representation rule violates the class definition and is a different theory rather than another PR solution.

The second-jet and effective-field-theory constructions displayed in the appendices make that last distinction explicit. They produce freedom only by dropping canonical normalization or by adding

independent operator data. They therefore have a named nonzero admissibility defect and cannot enlarge the PR quotient. A determinant expression satisfying the same normalized response conditions is an equivalent microscopic representation of the already unique D_3 operator, not an additional coefficient or branch.

Direct postulate derivation and canonical logical closure are the two accepted provenance routes. Neither route imports empirical targets, and neither adds an adjustable postulate. The completed class has no unresolved selector and requires no residual ranking to choose among equations. PR-IV therefore closes the mathematical question posed here: under the developed PR guidelines, all admissible formulations belong to one equation-equivalence class and no other PR solution architecture is allowed.

A Base-postulate Satisfaction and Canonical-closure Selection

This appendix verifies the layered structure of the PR-I construction. Let T_{PR} denote the seven-postulate foundation and let the completed compact class additionally include the signed pre-projection boundary-incidence rule and the minimal required-channel projector rule proved in this paper. Fix $R_A > 0$ and $\Lambda_X^2 > 0$ and take the common foundational data

$$\Psi : \mathbb{R}^{1,3} \times (\mathbb{R}_w \times S_\theta^1) \rightarrow \mathbb{C}, \quad (55)$$

$$ds_{\text{int}}^2 = dw^2 + R_A^2 d\theta^2, \quad (56)$$

$$\mathcal{H}_P = L^2(\mathbb{R}_w \times S_\theta^1, R_A dw d\theta), \quad (57)$$

$$O_{\text{int}} = O_X + O_\theta, \quad (58)$$

$$O_X = -\frac{d^2}{dw^2} + \Lambda_X^2 \left(1 + w^2 + \frac{3}{4}w^4\right), \quad (59)$$

$$O_\theta = -R_A^{-2} \frac{d^2}{d\theta^2}. \quad (60)$$

The standard Schrödinger and periodic domains make the two operators self-adjoint [7], and quartic confinement gives a simple discrete radial spectrum.

For purposes of exhaustive comparison, define

$$\mathcal{M}_{\text{PR}} = (-1, \Pi_{13}), \quad \mathcal{M}_0 = (0, \Pi_{13}), \quad \mathcal{M}_{15} = (-1, \Pi_{15}), \quad (61)$$

where the first entry is σ_8 in

$$N_\sigma = 1 + T_A^3 + T_A^4 + T_X(T_A^5 + \sigma_8 T_A^8). \quad (62)$$

These three formal expansions share the same foundational data; the canonical closure equations determine which expansion belongs to the completed PR class.

Table 2: Layered satisfaction of the PR-I foundation and compact closure.

Condition	$\mathcal{M}_{\text{PR}}$	$\mathcal{M}_0$	$\mathcal{M}_{15}$	Selection witness
P1 through P6	Pass	Pass	Pass	Identical master field, internal geometry, Hilbert space, separable operator, and radial potential.
P7 projection setting	Pass	Pass	Pass	All five projection sectors are represented.
Signed boundary incidence	Pass	Nonzero defect	Pass	$ 1 + \sigma_8 = 0$ fixes $\sigma_8 = -1$.
Minimal required channels	Pass	Pass	Nonzero defect	Rank two together with the required ϕ_1, ϕ_3 channels fixes Π_{13} .

The exact boundary witness is

$$c_{-1} = \frac{3354902985}{4211081216}, \quad c_0 = \frac{52420359}{65798144}, \quad c_{-1} - c_0 = \frac{9}{4211081216} \neq 0. \quad (63)$$

For the spectral slices, the compressed radial traces are $\lambda_1 + \lambda_3$ and $\lambda_1 + \lambda_5$. Simplicity of the radial spectrum gives $\lambda_3 \neq \lambda_5$, and compressed trace is invariant under an allowed basis change. The formal comparison expansions are therefore distinct before the canonical closure conditions are applied.

Theorem A.1 (Axiom-by-axiom Uniqueness in the Completed Compact Class). *Among expansions of the displayed PR-I foundational data that satisfy signed pre-projection zero incidence and the minimal required-channel rank-two condition, the sole equivalence class is $\mathcal{M}_{\text{PR}} = (-1, \Pi_{13})$.*

Proof. Signed boundary incidence gives $1 + \sigma_8 = 0$ and hence $\sigma_8 = -1$. The minimal rank-two projector containing the required nondegenerate channels ϕ_1 and ϕ_3 has range $\text{span}\{\phi_1, \phi_3\}$ and is therefore Π_{13} modulo an allowed normalized basis representation. Both conditions have one zero, so the completed quotient contains only $[\mathcal{M}_{\text{PR}}]$. $\square$

This axiom-by-axiom check confirms that the postulates and canonical closure rules play complementary roles: the former supply the foundation, while the latter make the compact construction mathematically complete and unique without introducing a fitted parameter or diagnostic input.

B Propagation of Alternative Equations to Locked Diagnostics

The following out-of-class controls verify that empirical-residual ranking is not used in the proof. We propagated representative comparison values through the same equations and applied the locked PR-III references only after generation. “Correct” in this appendix means inside the diagnostic interval accepted by PR-III; it does not mean exact equality.

Table 3: Compact-branch propagation to the locked rubidium $\alpha^{-1} = 137.035999206(11)$ diagnostic.

Branch	Generated α^{-1}	Distance
Released preprint $(-1, \Pi_{13})$	137.035999207530239	0.1391σ
Alternative $(0, \Pi_{13})$	137.035999207470098	0.1336σ
Alternative $(+1, \Pi_{13})$	137.035999207409986	0.1282σ
Comparison $(-1, \Pi_{15})$	254.208754866216196	outside interval
Comparison $(-1, \Pi_{17})$	384.885897406270487	outside interval

The three Π_{13} branches are exactly distinct, but all lie inside the same one-standard-deviation interval. They are statistically indistinguishable at this gate, so residual ordering is deliberately not used. The exact signed boundary-incidence theorem supplies the selection.

For the PR-III electromagnetic correction, none of the 21 alternative bounded-degree monomials passes the rubidium one-sigma gate when used as a single unit-coefficient replacement term. The PR-III $D_1 + D_2$ sum does pass. This numerical result is corroborative only. The exponent rule is a fixed inherited PR-III finite-tier composition rule, and the 21 replacement monomials lie outside that declared grammar.

The electroweak coefficient propagation uses

$$\Pi_W = \frac{a}{N_{\text{gen}}} \epsilon_V, \quad \Pi_Z = \frac{b}{N_{\text{gen}}} \epsilon_V + \frac{k}{N_{\text{gen}}} \epsilon_V^2. \quad (64)$$

Besides the released-preprint $(a, b, k) = (2, -1, 2)$, two alternatives pass both locked one-sigma mass gates:

Table 4: Electroweak coefficient triplets passing both locked mass gates.

(a, b, k)	M_W (GeV)	M_Z (GeV)	M_W distance	M_Z distance
$(2, -1, 1)$	80.37394285	91.18590923	0.3566σ	0.8051σ
$(2, -1, 2)$ released preprint	80.37394285	91.18767643	0.3566σ	0.0364σ
$(2, -1, 3)$	80.37394285	91.18944360	0.3566σ	0.8779σ

All three coefficients lie inside the stated diagnostic gate, so that gate is corroborative rather than generative. The canonical normalized-response theorem supplies the exact selection $k = 2$ independently of residual ranking.

C Locked Candidate-class Verification

The out-of-class comparison values were next subjected to every presently applicable locked constraint. A comparison is classified by an already-fixed physical ledger or by a diagnostic whose candidate set and tolerance were fixed independently. Residual ranking is not an equation-selection rule.

The audit contains 103 candidate-test findings. Eighty-four lie outside physical diagnostic intervals, one has a nonzero exact-ledger defect, one is conditional, nine tests are nondiscriminating, and eight remain numerically nondiscriminated. The classification is dominated by 61 electroweak coefficient triplets, 21 alternative single-monomial electromagnetic replacements, and the two nonminimal radial projectors Π_{15} and Π_{17} .

C.1 Compact Branches

The fine-structure diagnostic places Π_{15} and Π_{17} outside its interval by many orders of magnitude. At fixed Π_{13} , however, all three terminal comparison coefficients $\sigma_8 = -1, 0, +1$ remain numerically nondiscriminated. The quasar observable

$$\Delta v_{\text{high-low}} = \text{median}(\Delta v_{\text{est}})_{\text{high L}} - \text{median}(\Delta v_{\text{est}})_{\text{low L}} \quad (65)$$

contains no σ_8 , c_{bc} , or Π dependence in its released-preprint form. Each sign branch therefore makes the same negative-sign prediction. This successful PR-I stress test supports their common reduct but cannot select among its expansions.

The Kerr residual bound [1] is also unchanged when only σ_8 is varied at fixed Π_{13} , because the radial spectral gap is unchanged. The magnetic law

$$B_{\text{geo}}^{\text{PR}} = \frac{(\hbar/e)(Z_A/c_{\text{bc}})}{A_{\text{proj}}^{(\theta)}} \quad (66)$$

does inherit the small branch dependence of c_{bc} . Its present use is an inverse area constraint, however, and the realized source area is not fixed independently. It consequently does not select a terminal sign without a separate measurement or derivation of $A_{\text{proj}}^{(\theta)}$.

C.2 PR-II and PR-III Branches

The vectorlike $t = 0$ hypercharge family has a nonzero defect under the fixed nonzero electromagnetic charge ledger. Complete generation replicas with $N_{\text{gen}} \neq 3$ do not match observed multiplicity once sheet multiplicity is identified with physical generation multiplicity, while every positive replication count satisfies the local-anomaly and Witten gates. The normalized complete/nonredundant realization theorem supplies the exact selection.

None of the 21 tested comparison single-monomial replacements for $D_1 + D_2$ lies inside the locked fine-structure gate. This classifies that finite candidate family without making a statement about arbitrary higher-order combinations. Of the 64 tested electroweak triplets, only

$$(a, b, k) = (2, -1, 1), \quad (2, -1, 2), \quad (2, -1, 3) \quad (67)$$

pass both locked mass gates. All three therefore pass the diagnostic gate. Residual ranking is not an admissibility rule, and the exact normalized charged-adjoint theorem rather than best fit selects $k = 2$ in the canonical response class.

The stress audit therefore supplies substantial physical discrimination. Appendices F and G now prove $\sigma_8 = -1$ and the representation factor $k = 2$ inside explicit boundary and normalized projected-adjoint classes. The complete real-symmetric gauge-index coefficient classification at the fixed D_3 Lorentz/momentum kernel simultaneously shows that the three broad structural gates alone retain every real k . The remaining branches are resolved by the exact boundary and canonical-response closure theorems. Those target-independent principles are defining conditions of the completed admissible class. A physical- D_3 Hessian satisfying them is an equivalent microscopic representation and cannot introduce another coefficient. The full matrix and reproducible classifications are provided with the source archive.

D Exact Closure and Equation Exhaustion for Three Selectors

The three values below have exact uniqueness arguments inside the completed PR realization classes fixed by the admissibility guidelines. Their theorem domains are stated explicitly and no diagnostic preference enters.

D.1 Terminal Orientation

Let $E = XA^8$ be an independent terminal word. In the signed finite-rank pre-projection exclusion class, the raw return grammar contributes one copy of E , the terminal counterterm contributes $\sigma_8 E$, and admissibility requires zero total E incidence. Exact coefficient extraction gives

$$[E](C_{\text{raw}} + N_\sigma) = 1 + \sigma_8 = 0, \quad \sigma_8 = -1. \quad (68)$$

Every other value leaves the nonzero defect $(1 + \sigma_8)E$. The independent-word grammar and pre-projection zero-incidence rule define the target-independent PR-I boundary closure; if a later projector merely annihilates E , the signs instead become quotient-equivalent. Appendix F states the exact theorem and its domain.

D.2 Neutral Quadratic Representation Factor

At one fixed D_3 Lorentz/momentum kernel and declared scalar order, an exact rank reduction over all ten components of its real symmetric gauge-index coefficient matrix on (W^1, W^2, Z, A) gives dimensions four after residual $U(1)$ invariance, three after requiring the charged sector unchanged, and one after photon nullity. The complete structural family within this fixed-kernel matrix class is therefore

$$D_3(k) = \frac{\epsilon_V^2}{N_{\text{gen}}} k P_Z, \quad k \in \mathbb{R}. \quad (69)$$

Thus the structural gates fix a one-dimensional subspace but not its scalar. Independent Lorentz/derivative structures remain outside this matrix classification. For the normalized adjoint matrices $(F_a)_{bc} = -i\epsilon_{abc}$ and charged projector $P_{\text{ch}} = \text{diag}(1, 1, 0)$, define

$$K_{ab} = \text{Tr}_{\text{ad}}(P_{\text{ch}} F_a P_{\text{ch}} F_b). \quad (70)$$

Direct contraction gives $K_{ab} = 2\delta_{a3}\delta_{b3}$. Hence, if the PR-III operator is exactly

$$(D_3)_{ab} = \frac{\epsilon_V^2}{N_{\text{gen}}} K_{ab} \quad (71)$$

with unit prefactor and no other same-order contribution, then its charged entries vanish and $D_{3,Z} = 2\epsilon_V^2/N_{\text{gen}}$. Thus $k = 2$ is unique in this contraction class. More strongly, Theorem G.5 proves that basis invariance, orthogonal-mode additivity, quadratic homogeneity, and unit-charge normalization uniquely force this trace contraction in the minimal response class. On the complete structural family the exact squared defect is $(\epsilon_V^2/N_{\text{gen}})^2(k-2)^2$, whose sole real zero is two. Dropping the unit-prefactor identity creates the outside-class family $D_\lambda = 2\lambda(\epsilon_V^2/N_{\text{gen}})P_Z$; every $\lambda \neq 1$ violates canonical unit normalization and is not a PR solution. The locked one-sigma mass gate permits $0.7684347876 \dots < k < 3.1450923435 \dots$, including the fixed alternatives $k = 1, 2\cos^2\theta_W$, and 3. Appendix G gives the necessary-and-sufficient four-defect candidate characterization. Its normalization defect is exactly the target-independent canonical response condition. A determinant Hessian with the same normalized physical- Z map is an equivalent microscopic realization of the unique operator.

D.3 Generation Identification

Let V_{br} be the three-dimensional broken electroweak tangent space and $\Phi_N : S_N \rightarrow V_{\text{br}}$ the terminal-sheet incidence map. Completeness makes Φ_N surjective and nonredundancy makes it injective. Rank-nullity then gives

$$N = \text{rank } \Phi_N + \dim \ker \Phi_N = 3 + 0 = 3. \quad (72)$$

Count rigidity alone does not identify every invertible 3×3 incidence matrix under normalized basis changes, because unequal singular values would remain. The minimal normalized-incidence equation class therefore also requires metric compatibility,

$$\Phi_N^\dagger \Phi_N = I_N, \quad (73)$$

which assigns unit response to each normalized sheet and uses no measured target. Once $N = 3$, every such map is orthogonal in the real realization (unitary after complexification) and is equivalent to I_3 under allowed normalized incidence-basis transformations. Completeness and nonredundancy therefore prove the count, while Eq. (73) closes the incidence equation itself without a continuous singular-value modulus. Consequently, $N < 3$ is incomplete and $N > 3$ has a redundant sheet combination in that realization class. The direct sheet map is the target-independent PR-II canonical closure used by the completed admissible class. Appendix E states the complete integer theorem.

D.4 Verdict

All three values are exactly rigid in their stated PR realization classes and none is selected by best fit. The signed-incidence, complete/nonredundant sheet, and canonical normalized-response principles are therefore non-fitted logical closures. The broader common-reduct classes explain why these principles are stated explicitly; they do not supply alternative solutions inside the completed classes. The framework-native motivation and theorem domain of each closure are part of the final class definition.

Theorem D.1 (Equation-level Exhaustion of the Three Closure Selectors). *For the declared signed terminal-boundary, normalized complete/nonredundant generation-incidence, and fixed-kernel minimal quadratic-response classes, every admissible equation is operator-equivalent to the corresponding canonical PR equation. Their unique normal-form values are*

$$\boxed{\sigma_8 = -1, \quad (N_{\text{gen}}, [\Phi_3]) = (3, [I_3]), \quad D_3 = 2\rho P_Z \quad (k = 2).} \quad (74)$$

Every inequivalent member of those candidate classes has a nonzero exact defect.

Proof. For the terminal boundary class, every candidate reduces in the independent word algebra to the fixed raw grammar plus

$$N_\sigma = 1 + T_A^3 + T_A^4 + T_X(T_A^5 + \sigma T_A^8), \quad \sigma \in \mathbb{R}. \quad (75)$$

All other coefficients are fixed, so the XA^8 coefficient is a complete normal-form invariant. The raw return contributes one copy of the independent word $E = XA^8$ and the cofactor contributes σE . The independently declared pre-projection exclusion rule therefore has defect

$$d_8(\sigma) = |[E](C_{\text{raw}} + N_\sigma)| = |1 + \sigma|, \quad (76)$$

whose sole real zero is $\sigma = -1$. Any algebraic rewriting related by a nowhere-singular multiplier is equivalent; every other coefficient leaves a nonzero forbidden incidence. If exclusion is instead defined only after a later projector annihilates E , the coefficients become quotient-equivalent, which is the separate theorem domain already identified above.

For generation incidence, write $r = \text{rank } \Phi_N$ for $\Phi_N : S_N \rightarrow V_{\text{br}}$ with $\dim V_{\text{br}} = 3$. On $0 \leq r \leq \min\{N, 3\}$, completeness and nonredundancy have the exact nonnegative defects

$$d_{\text{comp}}(N, r) = 3 - r, \quad d_{\text{red}}(N, r) = N - r. \quad (77)$$

Both vanish if and only if $(N, r) = (3, 3)$. This fixes the count but, under normalized basis changes, does not alone fix every invertible incidence matrix. The declared equation class also contains the target-independent metric-compatibility defect

$$d_{\text{norm}}(\Phi_3) = \|\Phi_3^\dagger \Phi_3 - I_3\|_F^2. \quad (78)$$

Its zero set consists of orthogonal incidence maps (unitary after complexification). Any two are related by normalized orthogonal or unitary changes of the sheet or broken-direction basis, so the complete normal form is $(N, [\Phi_3]) = (3, [I_3])$. Without this normalization, the theorem proves only the count $N = 3$, not a unique incidence operator.

For the neutral response, Theorem G.1, charged-sector preservation, and photon nullity exhaust every fixed-kernel real symmetric candidate as

$$D_3(k) = \rho k P_Z, \quad k \in \mathbb{R}. \quad (79)$$

The corresponding residual equation is $E_k(D) = D - \rho k P_Z = 0$; thus this application has the same residual-map form as Theorem 3.1. The normalized eigenvalue multiset of D_3/ρ makes k a complete equivalence invariant. Theorem G.5 exhausts every response functional satisfying basis invariance, orthogonal-mode additivity, quadratic homogeneity, and unit-charge normalization and gives $\mathcal{F}(q) = \text{Tr}(q^\dagger q)$. On the charged-adjoint spectrum $(+1, -1)$, $\mathcal{F}(q) = 2$, and the complete defect is

$$d_k(k) = \|D_3(k) - 2\rho P_Z\|_F^2 = \rho^2(k - 2)^2. \quad (80)$$

Its sole real zero is $k = 2$. A differently written formula that reduces to the same trace on the complete declared charge-operator domain is equivalent. A formula that merely returns two on this single matrix but differs on another admissible charge operator is not the same response law: it either lies outside the declared response-functional domain or has a nonzero defect under at least one of the four functional conditions.

Each selector satisfies Theorem 3.1. The composition is triangular, not an assumption of three independent Cartesian factors: terminal incidence first fixes $\sigma_8 = -1$; normalized sheet incidence fixes $N_{\text{gen}} = 3$ and hence the inherited $\rho = \epsilon_V^2/3$; on that unique fiber, the response defect fixes $k = 2$. The combined dependency-ordered normal form therefore has the sole zero $(-1, 3, [I_3], 2)$, proving the result. $\square$

Proposition D.2 (Completed Equation-coordinate Quotient). *Let Θ_{closed} be the dependency-ordered normal-form space of the explicitly declared minimal radial, minimal coordinate-projector, signed terminal-boundary, nonzero-ledger hypercharge, fixed outer-product photon, complete/nonredundant normalized generation-incidence, and fixed-kernel normalized-response selector equations audited in this paper. Inherited data are frozen before a downstream fiber is classified, and $\sim_{\text{sel}}$ applies the stated ledger-preserving equivalence in each fiber. Then*

$$\{\theta \in \Theta_{\text{closed}} : d_{\text{closed}}(\theta) = 0\} / \sim_{\text{sel}} = \{[\theta_{\text{PR}}]_{\text{sel}}\}. \quad (81)$$

Thus another in-class equation that works is either a ledger-preserving representation of the same physical operator or has a nonzero defect. These are all adjustable equation-selection coordinates of $\mathfrak{A}_{\text{PR}}^{\text{can}}$; the remaining public sector maps are fixed inherited relations. This proposition is therefore the selector-coordinate factor used in the global singleton Theorem 8.1.

Table 5: Complete Normal Forms in the Completed Selector-coordinate Quotient.

Block	Normal form modulo stated equivalence	Unique-zero certificate
Radial	(a_2, a_4) in the minimal even quartic	$(a_2 - 1)^2 + (a_4 - 3/4)^2 = 0$
Coordinate projector	$\text{Ran } \Pi$ in the minimal required-channel rank-two class	rank two and inclusion of ϕ_1, ϕ_3 force $\text{Ran } \Pi = \text{span}\{\phi_1, \phi_3\}$
Terminal boundary	$\sigma_8 \in \mathbb{R}$	$ 1 + \sigma_8 = 0$
Hypercharge	anomaly-solution orbit $[Y]$	the factored anomaly ideal plus the fixed nonzero charge ledger leaves one orbit modulo normalization and up/down relabeling
Photon	normalized null line $[v_\gamma]$	the fixed rank-one outer-product neutral matrix has a one-dimensional kernel
Generation incidence	$(N, [\Phi_N])$ in the normalized incidence class	completeness, nonredundancy, and $\ \Phi_3^\dagger \Phi_3 - I_3\ _F^2 = 0$ leave $(3, [I_3])$
Neutral response	$(k, [\mathcal{F}])$ at fixed kernel	the four response conditions give $\mathcal{F} = \text{Tr}(q^\dagger q)$ and $\rho^2(k - 2)^2 = 0$

Proof. The complete normal form and unique-zero condition for each closed block are listed in Table 5 and proved in the cited sector sections and appendices. Order the blocks by the dependency ledger. The first block has one admissible orbit. Inductively, freeze its unique inherited data and apply the next block's complete normal-form theorem on that fiber; it too has one admissible orbit. Repeating this finite argument through the neutral-response fiber gives one dependency-compatible selector orbit. No Cartesian independence assumption is used. $\square$

E Generation-count Rigidity in the PR-II Sheet Class

The generation ambiguity can be closed inside the generation-sheet class actually used by PR-II. Let

$$G = SU(2)_L \times U(1)_Y, \quad H = U(1)_{\text{em}}, \quad V_{\text{br}} = T_{[eH]}(G/H). \quad (82)$$

The broken-direction space has the exact dimension

$$\dim V_{\text{br}} = \dim G - \dim H = (3 + 1) - 1 = 3. \quad (83)$$

Let S_N be the span of N distinct terminal generation sheets, and let

$$\Phi_N : S_N \longrightarrow V_{\text{br}} \quad (84)$$

be their broken-direction incidence map. The direct mapping described in PR-II has two structural requirements. Completeness requires Φ_N to be surjective, so no broken electroweak direction is omitted. Non-redundancy requires Φ_N to be injective, so no nonzero sheet combination has a zero geometric image.

Theorem E.1 (PR-II generation-count rigidity). *A complete and nonredundant terminal-sheet realization exists if and only if $N = 3$. Consequently $N_{\text{gen}}^{\text{PR}} = 3$ is the unique count in this construction class.*

Proof. Surjectivity implies $\text{rank } \Phi_N = \dim V_{\text{br}} = 3$, hence $N \geq 3$. Injectivity implies nullity $\Phi_N = 0$. Rank-nullity then gives

$$N = \dim S_N = \text{rank } \Phi_N + \text{nullity } \Phi_N = 3 + 0 = 3. \quad (85)$$

Conversely, a basis of S_3 can be mapped to a basis of V_{br} , giving an isomorphism. Therefore three is necessary and sufficient. $\square$

This proof exhausts every positive integer. If $N < 3$, the map has rank at most N and cannot cover the quotient. If $N > 3$, every surjective map has nullity at least $N - 3$, producing redundant or geometrically ungrounded sheet combinations. Thus every alternative count breaks either completeness or nonredundancy.

The theorem domain is PR-II's declared terminal-sheet realization, which directly maps the replicated matter ledger across the broken compact projection directions. Within that complete domain, every $N \neq 3$ has a strict completeness or redundancy defect and is not a PR solution. No measured generation count, mass, mixing angle, or diagnostic residual enters the proof.

The closure principle is the minimal faithful realization of the broken compact space: surjectivity represents every broken direction and injectivity excludes an unobservable or redundant sheet combination. These conditions are logical, target-independent, and contain no adjustable coefficient. They constitute the canonical PR-II closure of the sheet-incidence map. Direct P1 through P7 ancestry and proved canonical closure are both accepted provenance routes under Proposition 2.1.

F Terminal-orientation Rigidity in the PR-I Boundary Class

The terminal-sign coordinate is solved exactly inside the signed finite-rank boundary class used by PR-I. Introduce a formal trace-length variable z and a one-boundary marker X , with $X^2 = 0$. The compact interior return series and terminal numerator are

$$I(z) = \frac{1}{1 - z^3 - z^4}, \quad N_\sigma(z, X) = 1 + z^3 + z^4 + X(z^5 + \sigma_8 z^8). \quad (86)$$

The complete pre-projection one-boundary series is

$$F_\sigma(z, X) = N_\sigma(z, X)I(z). \quad (87)$$

Let $E = Xz^8$ be a nonzero independent word. There are two separate ledgers. The raw recurrence $Xz^5[z^3]I$ contributes $[E]C_{\text{raw}} = 1$, while the explicit cofactor counterterm contributes $[E]N_{\sigma, \text{corr}} = \sigma_8$. Thus the released-preprint $-Xz^8$ remains in the numerator precisely to cancel the raw positive incidence in the full series.

The finite-rank orientation rule excludes the first repeated length-eight recirculation before projection. In the formal return algebra, this is the exact incidence equation

$$[Xz^8]F_\sigma = 0. \quad (88)$$

Theorem F.1 (Terminal-orientation Rigidity). *Equation (88) has the unique real solution $\sigma_8 = -1$. Hence every other integer or real coefficient has nonzero PR-I finite-rank boundary-exclusion defect $|1 + \sigma_8|$.*

Proof. The interior series begins

$$I(z) = 1 + z^3 + z^4 + z^6 + 2z^7 + z^8 + \mathcal{O}(z^9). \quad (89)$$

Only $Xz^5[z^3]I$ and $\sigma_8 Xz^8[z^0]I$ contribute to $[Xz^8]F_\sigma$. Both interior coefficients equal one, so

$$[Xz^8]F_\sigma = 1 + \sigma_8. \quad (90)$$

The exclusion condition gives $1 + \sigma_8 = 0$, and therefore $\sigma_8 = -1$ uniquely. $\square$

For $\sigma_8 = 0$, one excluded length-eight return remains. For $\sigma_8 = +1$, its multiplicity is two. More generally every $\sigma_8 \neq -1$ leaves nonzero incidence $1 + \sigma_8$. The exclusion uses no fine-structure value or other diagnostic and is not a best-fit selection.

As with Appendix E, the theorem domain must be stated. The result closes the coefficient inside PR-I's signed pre-projection boundary accounting. Equation (88) need not appear verbatim in P1–P7; it is admissible as a target-independent closure principle when the finite-rank construction excludes the first repeated recirculation before projection. Under that declared interpretation, $\sigma_8 = -1$ is uniquely necessary. If “exclusion” were instead defined only by a later projection that annihilates E , every value of σ_8 would become quotient-equivalent. The manuscript therefore states explicitly that signed pre-projection accounting is the intended PR construction.

G Complete fixed-kernel D_3 Gauge-index Classification and Canonical Reduction

Set

$$\rho = \frac{\epsilon_V^2}{N_{\text{gen}}} > 0. \quad (91)$$

This appendix classifies every real symmetric gauge-index coefficient matrix multiplying the fixed Lorentz/momentum kernel at the declared scalar order. Independent Lorentz structures, CP-odd or complex Hermitian kernels, higher derivatives, nonlocal operators, and additional fields violate that fixed operator inventory and belong to different theory classes. Algebraically equivalent equations necessarily describe the same tensor.

Throughout this appendix the reproducibility source map references repository commit

9680f70b21f04df0350fd857471b1aff4ec07ead.

The corresponding DOI manuscripts and the current PR-III manuscript supply the inherited representation data, determinant architecture, and finite-tier construction used here. The source-version record is retained solely for reproducibility and leaves every PR-I through PR-III result intact.

Let

$$V_{\text{EW}} = \text{span}_{\mathbb{R}}\{W^1, W^2, W^3, B\} = \mathcal{C} \oplus \mathcal{N}, \quad \mathcal{C} = \text{span}\{W^1, W^2\}. \quad (92)$$

The canonical real inner product makes (W^1, W^2, W^3, B) orthonormal. For a unit vector v , let $P_v = v \otimes v$ be its orthogonal projector. Then

$$\|P_v\|_F^2 = \text{Tr}(P_v^2) = 1, \quad P_{\text{ch}} = P_{W^1} + P_{W^2}. \quad (93)$$

PR-II fixes the physical neutral basis [4]

$$Z = cW^3 - sB, \quad A = sW^3 + cB, \quad c = \cos \theta_W, \quad s = \sin \theta_W. \quad (94)$$

G.1 Exhaustion under residual symmetry

In the ordered basis (W^1, W^2, Z, A) , residual $U(1)_{\text{em}}$ acts as

$$R(\alpha) = \text{diag}(R_2(\alpha), 1, 1), \quad J = R'(0) = \text{diag}(J_2, 0, 0), \quad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (95)$$

Theorem G.1 (Complete residual- $U(1)$ classification). *For $D \in \text{Sym}(V_{\text{EW}})$, the identity*

$$R(\alpha)^T D R(\alpha) = D \quad \text{for every } \alpha \in \mathbb{R} \quad (96)$$

holds if and only if

$$D = aP_{\text{ch}} + zP_Z + x(Z \otimes A + A \otimes Z) + \gamma P_A \quad (97)$$

for arbitrary $a, z, x, \gamma \in \mathbb{R}$.

Proof. Write the symmetric block matrix

$$D = \begin{pmatrix} C & L \\ L^T & N \end{pmatrix}, \quad C, N \in \text{Sym}_2(\mathbb{R}). \quad (98)$$

Differentiating Eq. (96) at $\alpha = 0$ gives $[D, J] = 0$, hence

$$CJ_2 = J_2C, \quad J_2L = 0, \quad L^TJ_2 = 0. \quad (99)$$

Because J_2 is invertible, $L = 0$. If $C = \begin{pmatrix} u & v \\ v & w \end{pmatrix}$, the first equation gives $v = 0$ and $u = w$, so $C = aI_2$. The residual group acts trivially on the neutral plane, leaving N as an arbitrary real symmetric 2×2 matrix. Expanding N in the (Z, A) basis gives exactly Eq. (97). Direct substitution proves the converse. $\square$

The accompanying exact rational row reduction starts with all ten independent components of a symmetric 4×4 matrix. Equation (96) has rank six and therefore the four-dimensional solution space in Theorem G.1. Imposing an unchanged charged sector,

$$P_{\text{ch}}DP_{\text{ch}} = 0, \quad (100)$$

sets $a = 0$; the combined rank is seven and the solution dimension is three. If D_3 is a zero-momentum mass-kernel correction, photon preservation adds

$$DA = 0. \quad (101)$$

Equation (97) then gives $x = \gamma = 0$. The combined rank is nine, and the complete structural family is therefore

$$D(k) = \rho k P_Z, \quad k \in \mathbb{R}. \quad (102)$$

Every real k satisfies residual symmetry, charged-sector preservation, and photon nullity. These conditions fix a one-dimensional subspace, not its scalar. For a momentum-dependent polarization, ordinary Lorentz transversality must not be silently replaced by Eq. (101); the displayed result uses the stronger mass-kernel condition relevant to the PR-III scalar mass correction.

The current local PR-III revision states the same fixed-kernel exhaustion. Its stable label is

`prop:priii-d3-structural-classification.`

The revision likewise records that the structural gates classify the projector direction but do not select the scalar k .

Theorem G.2 (Frozen-reduct Expansions). *Let $\mathcal{L}_0$ be the target-free language containing the symbols of P1–P7 together with the frozen background objects used above— ρ , $V_{\text{EW}} = \mathbb{R}^4$ with its canonical metric and charge grading, real c, s with $c^2 + s^2 = 1$, and the physical Z, A basis and residual action—but no D_3 tensor symbol. Let T_0 be exactly P1 through P7 together with those enumerated target-free background axioms. It contains neither the candidate equation $D_{3,Z} = 2\rho$ nor downstream diagnostic comparisons, and no additional target-free PR-II/III sentence is silently included. Extend the language by a real symmetric symbol D_3 , and set*

$$\mathsf{T}_{\text{struct}} = \mathsf{T}_0 \cup \left\{ \begin{array}{l} R(\alpha)^T D_3 R(\alpha) = D_3 \quad (\forall \alpha \in \mathbb{R}), \\ P_{\text{ch}} D_3 P_{\text{ch}} = 0, \quad D_3 A = 0 \end{array} \right\}. \quad (103)$$

The theory T_0 is nonempty. For every $k \in \mathbb{R}$, the expansion of the explicit model $\mathfrak{M}_0$ constructed below,

$$\mathfrak{M}_k = (\mathfrak{M}_0, D_3 = \rho k P_Z) \quad (104)$$

models $\mathsf{T}_{\text{struct}}$. Moreover, $\mathfrak{M}_k$ and $\mathfrak{M}_{k'}$ are inequivalent under normalized orthogonal basis changes whenever $k \neq k'$.

Proof. Start with the explicit P1–P7 common reduct constructed in Appendix A. Adjoin the independent target-free background $V_{\text{EW}} = \mathbb{R}^4$ with orthonormal basis (W^1, W^2, W^3, B) ; choose $\rho > 0$ and $c^2 + s^2 = 1$; and define Z, A and $R(\alpha)$ by Eqs. (94) and the residual action above. These definitions alter no P1–P7 symbol and contain no D_3 equation, so by the explicit definition of T_0 they give an explicit $\mathfrak{M}_0 \models \mathsf{T}_0$. Changing the interpretation of the new D_3 symbol cannot alter any $\mathcal{L}_0$ sentence, so every expansion preserves all P1 through P7 and the enumerated target-free background axioms. Equation (102) verifies each added structural sentence. Finally,

$$\text{spec}_{\text{mult}}(D_3/\rho) = \{k, 0, 0, 0\}. \quad (105)$$

Normalized orthogonal similarity preserves this eigenvalue multiset, proving inequivalence when $k \neq k'$. $\square$

This is the axiom-by-axiom common-reduct proof of an uncountable structural model class. If Eq. (101) is not a derived mass-kernel condition, that class is larger still.

G.2 Canonical Adjoint Trace

PR-II fixes [4]

$$T_a = \frac{\sigma_a}{2}, \quad \text{Tr}(T_a T_b) = \frac{1}{2} \delta_{ab}, \quad [T_a, T_b] = i \epsilon_{abc} T_c, \quad (106)$$

and places the weak projection modes in the adjoint $j = 1$ representation. On $\mathcal{H}_{\text{ch}} = \text{span}\{T^+, T^-\}$, $[Q, T^+] = +T^+$ and $[Q, T^-] = -T^-$. Let $e_3 = (0, 0, 1)^T$ and $P_3 = e_3 \otimes e_3$. Define

$$(F_a)_{bc} = -i \epsilon_{abc}, \quad P_{\text{ch}}^{\text{ad}} = \text{diag}(1, 1, 0), \quad K_{ab} = \text{Tr}_{\text{ad}}(P_{\text{ch}}^{\text{ad}} F_a P_{\text{ch}}^{\text{ad}} F_b). \quad (107)$$

Here P_3 denotes the adjoint third-axis projector and is unrelated to the base postulate carrying the label P3.

Lemma G.3 (Exact projected-adjoint tensor). *With the frozen normalization in Eq. (106),*

$$K_{ab} = 2\delta_{a3}\delta_{b3} = 2(P_3)_{ab}. \quad (108)$$

Proof. $P_{\text{ch}}^{\text{ad}} F_1 P_{\text{ch}}^{\text{ad}}$ and $P_{\text{ch}}^{\text{ad}} F_2 P_{\text{ch}}^{\text{ad}}$ vanish, whereas

$$P_{\text{ch}}^{\text{ad}} F_3 P_{\text{ch}}^{\text{ad}} = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \text{Tr}[(P_{\text{ch}}^{\text{ad}} F_3 P_{\text{ch}}^{\text{ad}})^2] = 2. \quad (109)$$

Thus only K_{33} is nonzero and it equals two. Equivalently, the charged eigenvalues of ad_Q are $(+1, -1)$, so their squared trace is $1^2 + (-1)^2 = 2$; or directly $\sum_{b,c} \epsilon_{3bc}^2 = \epsilon_{312}^2 + \epsilon_{321}^2 = 2$. $\square$

The charged scalar trace is invariant under normalized transformations that preserve Q and the charged subspace. The displayed component tensor is covariant relative to the distinguished neutral axis. The lemma fixes a canonical representation trace; it does not fix the dynamical scalar multiplying an effective operator.

This exact representation result is now also stated in the current local PR-III revision under the stable label

`prop:priii-exact-charged-adjoint-trace.`

G.3 Canonical logical closure of k

Let

$$q = \text{ad}_Q|_{\mathcal{H}_{\text{ch}}}, \quad \mathcal{H}_{\text{ch}} = \text{span}\{T^+, T^-\}. \quad (110)$$

The frozen charge normalization gives $qT^+ = T^+$ and $qT^- = -T^-$. Hence

$$q^\dagger q = I_{\mathcal{H}_{\text{ch}}}, \quad \text{Tr}_{\text{ch}}(q^\dagger q) = 2. \quad (111)$$

Proposition G.4 (Exact canonical-response value). *If the physical neutral correction is the canonically normalized quadratic charged-adjoint response*

$$D_{3,Z} = \rho \text{Tr}_{\text{ch}}(q^\dagger q), \quad (112)$$

then

$$D_{3,Z} = 2\rho, \quad k = 2. \quad (113)$$

This value is invariant under every unitary change of charged basis that preserves the normalized charge operator. It contains no empirical target, phase choice, continuous coefficient, or residual ranking.

Proof. Equation (111) substituted into Eq. (112) gives the result. Under a unitary charged-basis change U , $q^\dagger q$ changes by similarity, so its trace remains two. $\square$

The response identity is not an arbitrary choice. It is forced inside the minimal quadratic-response class by the following functional theorem.

Theorem G.5 (Uniqueness of the normalized quadratic response). *Let $\mathcal{F}$ assign a nonnegative real number to every normal finite-dimensional charge operator. For every dimension-compatible unitary U and every $a \in \mathbb{C}$, suppose*

$$\begin{aligned} \mathcal{F}(UqU^\dagger) &= \mathcal{F}(q), & \text{unitary-basis invariance,} \\ \mathcal{F}(q_1 \oplus q_2) &= \mathcal{F}(q_1) + \mathcal{F}(q_2), & \text{orthogonal-mode additivity,} \\ \mathcal{F}(aq) &= |a|^2 \mathcal{F}(q), & \text{quadratic homogeneity,} \\ \mathcal{F}([1]) &= 1, & \text{unit-charge normalization.} \end{aligned} \quad (114)$$

Then, uniquely,

$$\mathcal{F}(q) = \text{Tr}(q^\dagger q). \quad (115)$$

Consequently the class of neutral corrections

$$D_3 = \rho \mathcal{F}(q) P_Z \quad (116)$$

contains one element for the PR charged-adjoint spectrum $(+1, -1)$, namely $D_3 = 2\rho P_Z$.

Proof. For a normal q , choose a unitary U such that $UqU^\dagger = \text{diag}(\lambda_1, \dots, \lambda_n)$. Unitary invariance and direct-sum additivity reduce $\mathcal{F}(q)$ to the sum of its one-dimensional values. Quadratic homogeneity and unit normalization give $\mathcal{F}([\lambda_i]) = |\lambda_i|^2$. Hence

$$\mathcal{F}(q) = \sum_i |\lambda_i|^2 = \text{Tr}(q^\dagger q). \quad (117)$$

Every step is forced, so no other functional satisfies all four conditions. For $q = \text{diag}(1, -1)$ the result is 2; substitution into Eq. (116) completes the proof. $\square$

The four hypotheses express the logical content of the PR extension: observables cannot depend on the charged basis, independent charged modes add in the spectral trace, the declared tier is quadratic in charge, and a single normalized mode has unit weight. They introduce neither a new scale nor an adjustable coefficient. The current local PR-III revision names Eq. (112) as $\mathbf{H}_Z$. It is therefore the canonical no-fit closure law. A complete neutral determinant satisfying the same conditions is an equivalent microscopic realization; it does not define another coefficient or branch.

There is an equivalent arithmetic observation in the released finite-tier ledger. The public formulas imply

$$D_{3,Z} = \epsilon_V \Pi_W^{C_2}, \quad \Pi_W^{C_2} = \frac{2}{N_{\text{gen}}} \epsilon_V, \quad (118)$$

and therefore $D_{3,Z} = 2\epsilon_V^2/N_{\text{gen}}$. This arithmetic identity is consistent with the canonical response theorem but is not used to select the response functional. Theorem G.5 provides the target-independent logical selection; an operator determinant is an equivalent realization when it produces the same normalized response.

G.4 Independent Determinant-Hessian Realization

The PR-III determinant architecture is instantiated for the residual electromagnetic background. A normalized physical- Z background family provides a complementary microscopic realization of the canonical response $\mathbf{H}_Z$. For such a realization, write the smooth multi-component blocks as $H_i[B^a]$, including the normalized physical neutral direction. This operator construction is an equivalent representation of the finite-tier canonical theorem. Write its block architecture as

$$\Gamma^{(1)}[B] = \sum_i \eta_i \text{Tr}'_i \log \left(H_i[B] H_i[0]^{-1} \right), \quad (119)$$

where the released weights are $\eta_{\text{scalar}} = \eta_{\text{vector}} = 1/2$ and $\eta_{\text{ghost}} = \eta_{\text{fermion}} = -1$. Set

$$G_i = H_i[0]^{-1}, \quad V_{i,a} = \left. \frac{\delta H_i}{\delta B^a} \right|_0, \quad W_{i,ab} = \left. \frac{\delta^2 H_i}{\delta B^a \delta B^b} \right|_0. \quad (120)$$

Lemma G.6 (Second Variation of the Reference-normalized Determinant). *Assume for each block that the primed subspace and operator domain are fixed, $H_i[B]$ is a twice differentiable invertible family there, and the relative variations are trace class in a topology permitting differentiation under the trace and cyclic rearrangement. Alternatively, for a zeta prescription assume that the same differentiated-trace identity holds without an additional local regularization anomaly. Then*

$$\left. \frac{\delta^2 \Gamma^{(1)}}{\delta B^a \delta B^b} \right|_0 = \sum_i \eta_i \text{Tr}'_i (G_i W_{i,ab} - G_i V_{i,a} G_i V_{i,b}). \quad (121)$$

Proof. For each block, $\delta \text{Tr} \log H = \text{Tr}(H^{-1} \delta H)$ and $\delta H^{-1} = -H^{-1}(\delta H)H^{-1}$. Differentiating once more and evaluating at $B = 0$ gives Eq. (121); cyclicity of the trace makes the expression symmetric in a, b . $\square$

A complete determinant realization specifies the neutral vertices $V_{i,a}$, seagulls $W_{i,ab}$, block weights and multiplicities, spectral trace, gauge-fixed vector-ghost-Goldstone organization [6], matter contribution, and the normalization that extracts the physical- Z self-energy from Eq. (121). These ingredients define an equivalent microscopic representation of the canonical-response theorem. If a selected regularization produces a local counterterm or anomaly, its contribution is recorded explicitly in the operator-basis and renormalization ledger rather than absorbed into k .

The following exact family characterizes the broader second-jet comparison domain and thereby distinguishes it from the completed minimal PR response class.

Theorem G.7 (Second-jet Completion Family). *Let $H_0 > 0$ satisfy the explicitly released Step-04 reference-block properties. Let X be a finite-rank spectral projector supported in the retained primed subspace, commuting with H_0 , and satisfying $\text{Tr}' X = 1$. Let z be a dimensionless canonically normalized coordinate along the physical Z background. For every $\xi \in \mathbb{R}$, define*

$$H_\xi[z] = H_0 \exp(\xi \rho z^2 X). \quad (122)$$

Then $H_\xi[0] = H_0$, every $H_\xi[z]$ is positive and reference-normalized, and its bosonic determinant contribution is

$$\frac{1}{2} \text{Tr}' \log (H_\xi[z] H_0^{-1}) = \frac{1}{2} \xi \rho z^2. \quad (123)$$

Consequently, its neutral Hessian is $\xi \rho$. This family parametrizes a deliberately broader second-jet comparison domain; it does not add PR solutions, because the completed minimal-response class also imposes the four normalized response conditions that uniquely give $k = 2$.

Proof. Commutation makes $H_\xi[z] H_0^{-1} = \exp(\xi \rho z^2 X)$. Functional calculus gives positivity for every real ξ , while the finite rank of X makes the relative trace finite. Taking the logarithm and using $\text{Tr}' X = 1$ gives the displayed quadratic action; differentiating twice gives $\xi \rho$. $\square$

This family is a scope comparison whose arbitrary second jet lies outside the canonical minimal-response class unless the complete normalized-response conditions are satisfied. The current PR-III manuscript records the corresponding construction at the stable label as a fixed kernel counter family. In particular, the special normalized block

$$H_{\text{ch}}[z] H_{\text{ch}}[0]^{-1} = \exp(\rho z^2 q^\dagger q) \quad (124)$$

would give $\Gamma^{(1)} = \rho z^2$ and hence $D_{ZZ} = 2\rho$, but Eq. (124) is one possible determinant realization of the canonical response identity.

G.5 Canonical Reduction to the Physical Tensor

Let

$$\mathcal{E} : \text{span}_{\mathbb{R}}\{P_3\} \longrightarrow \text{span}_{\mathbb{R}}\{P_Z\} \quad (125)$$

be a specified normalized real-linear isometry. The needed hypotheses are:

- H1. the declared minimal order- ρ neutral-response class contains no independent Wilson coefficient or additional same-order operator;
- H2. its scalar response obeys the basis-invariance, mode-additivity, quadratic-homogeneity, and unit-charge normalization conditions of Theorem G.5; together with H3 these conditions give $D = \rho \mathcal{E}(K)$ with unit mode weight;
- H3. $\mathcal{E}$ is real-linear and $\mathcal{E}(P_3) = P_Z$, as a normalized neutral-line map rather than an unannounced ordinary W^3 component projection;

H4. generator, trace, field, kinetic, ϵ_V , and generation normalizations are frozen; and

H5. Eqs. (96)–(101) hold.

Theorem G.8 (Canonical D_3 reduction). *Under H1–H5,*

$$D = 2\rho P_Z, \quad D_{3,W} = 0, \quad D_{3,Z} = \frac{2}{N_{\text{gen}}} \epsilon_V^2, \quad (126)$$

so $k = 2$ is the unique coefficient in this operator class.

Proof. Lemma G.3, H2, and H3 give

$$D = \rho \mathcal{E}(K) = 2\rho \mathcal{E}(P_3) = 2\rho P_Z. \quad (127)$$

H1 excludes a second same-order tensor, and H4 excludes absorbing a multiplier into the fields or ϵ_V . $\square$

This theorem uses no experimental residual and is not a best-fit selection. H2 is derived as the unique normalized quadratic functional in Theorem G.5; it introduces no adjustable coefficient. A gauge-fixed determinant calculation satisfying H1 through H5 provides the same map and cannot create an additional admissible normalization.

The current local PR-III revision records the corresponding reduction under the stable legacy label

`thm:priii-conditional-k-two.`

Within the canonical response class $\mathcal{H}_Z$, division by $\rho_{\text{PR}} > 0$ selects $k = 2$ uniquely and leaves no free coefficient.

G.6 Necessary Sufficient Candidate Characterization

For arbitrary $D \in \text{Sym}(V_{\text{EW}})$, define

$$\begin{aligned} E_{\text{sym}}(D) &= \sup_{\alpha} \|R(\alpha)^T D R(\alpha) - D\|_F, \\ E_{\text{ch}}(D) &= \|P_{\text{ch}} D P_{\text{ch}}\|_F, \\ E_{\gamma}(D) &= \|D A\|_2, \\ E_{\text{norm}}(D) &= |Z^T D Z - 2\rho|. \end{aligned} \quad (128)$$

Theorem G.9 (Complete Candidate Characterization). *All four defects in Eq. (128) vanish if and only if $D = 2\rho P_Z$.*

Proof. $E_{\text{sym}} = 0$ gives the complete family in Theorem G.1. Then $E_{\text{ch}} = 0$ sets $a = 0$, and $E_{\gamma} = 0$ sets $x = \gamma = 0$, leaving $D = z P_Z$. $E_{\text{norm}} = 0$ finally gives $z = 2\rho$. Direct substitution proves the converse. $\square$

Thus every other real symmetric tensor either has a nonzero structural defect or differs from the canonical response normalization. On the structural family (102),

$$D(k) - D(2) = \rho(k - 2)P_Z, \quad \|D(k) - D(2)\|_F^2 = \rho^2(k - 2)^2, \quad (129)$$

whose sole real zero is $k = 2$. The logical role of $E_{\text{norm}} = 0$ is now explicit: Theorem G.5 derives it as the unique target-independent functional in the declared minimal response class. A determinant derivation at the microscopic operator level is an equivalent realization and is not an additional condition of the uniqueness theorem.

G.7 Exclusion of Incomplete Structural Reductions and the Electroweak Mixing-normalization Distinction

If H2 is omitted, the resulting outside-class scalar family is

$$D_\lambda = \rho\lambda\mathcal{E}(K) = 2\lambda\rho P_Z, \quad \lambda \in \mathbb{R}, \quad (130)$$

passes the first three defects. Fixed rational choices require no fitted continuous parameter: $\lambda = 1/2$ gives $k = 1$, and $\lambda = 3/2$ gives $k = 3$. They show that the three broad structural gates alone do not determine the normalization. They are not proposed PR branches and do not satisfy the unit-charge normalization of the canonical response class.

The three linear structural gates in Theorem G.2 permit every real k . An independent strict positivity condition on the neutral correction would restrict the family to $k > 0$, while positive semidefiniteness would give $k \geq 0$. Either case remains uncountable and retains the fixed alternatives $k = 1, 2\cos^2\theta_W, 2$, and 3; neither selects two.

Checking only the stated ZZ scalar is still weaker. Every tensor

$$\rho\{2P_Z + x(Z \otimes A + A \otimes Z) + \gamma P_A\} \quad (131)$$

has the same ZZ entry for arbitrary x, γ . Photon nullity eliminates these two directions, but leaves the scalar family (130).

The ordinary electroweak rotation creates a separate normalization issue. In the (Z, A) basis,

$$2P_{W^3} = 2 \begin{pmatrix} c^2 & cs \\ cs & s^2 \end{pmatrix}. \quad (132)$$

Its ZZ pullback is $2c^2$, but it has ZA mixing and a photon component. A photon-safe tensor with the same scalar is $2c^2P_Z$, which is a different operator with nonzero canonical-normalization defect. The current PR-III revision records this same normalization distinction at the stable label `eq:priii-w3-to-physical-neutral-map`.

Proposition G.10 (Mixing-normalization Incompatibility). *Assume D is real symmetric, $0 < c < 1$, and $DA = 0$. It is impossible to have both $D_{W^3W^3} = 2\rho$ and $D_{ZZ} = 2\rho$.*

Proof. Since $W^3 = cZ + sA$, symmetry and $DA = 0$ give

$$D_{W^3W^3} = (cZ + sA)^T D (cZ + sA) = c^2 D_{ZZ}. \quad (133)$$

The two proposed nonzero equalities therefore imply $c^2 = 1$, a contradiction. $\square$

Consequently, three inequivalent maps give, respectively,

$$\begin{array}{ll} P_3 \mapsto P_Z : & D = 2\rho P_Z, \quad k = 2, \\ \text{component-preserving photon completion :} & D = (2\rho/c^2)P_Z, \quad k = 2/c^2, \\ \text{literal } W^3 \text{ embedding :} & D = 2\rho P_{W^3}, \quad D_{ZZ}/\rho = 2c^2. \end{array} \quad (134)$$

The declared PR map is $\mathcal{E}(P_3) = P_Z$, with canonical unit normalization. The literal $2P_{W^3}$ tensor and the photon-safe $2c^2P_Z$ tensor are distinct cases: the first has nonzero photon-nullity defect, while the second changes the normalization.

G.8 Locked Diagnostics do not Isolate the Coefficient

The released arithmetic is

$$M_Z(k) = M_{Z,C1} \sqrt{1 + \Pi_{Z,C2} + \rho k}. \quad (135)$$

Monotonicity of the positive square root gives the exact released strict one-standard-deviation open interval

$$0.768434787688685769 \dots < k < 3.145092343548119919 \dots \quad (136)$$

It is uncountable and its integer members are exactly $\{1, 2, 3\}$. Its endpoints have unit diagnostic distance and lie on the excluded boundary of the released strict acceptance gate. The exact diagnostic central-value solution is $k = 1.956749882341106 \dots$, not two. Representative results are

Table 6: Parameter-free D_3 normalizations under the locked PR-III mass diagnostic.

k	$M_Z(k)$ (GeV)	absolute diagnostic distance (σ)
1	91.1859092277010	0.80513
$2 \cos^2 \theta_W = 1.53728847 \dots$	91.1868587299446	0.35299
2	91.1876764310439	0.03640
3	91.1894436001394	0.87790

M_W and G_F are independent of k in the released generator. Thus the locked diagnostics are corroborative rather than generative. The exact normalized-response theorem, independent of best-fit ranking, fixes $k = 2$.

G.9 Source Mapping, PR-III Integration, and Exact Theorem

The PR-III construction supplies the representation data, the stated finite-tier candidate, and the reference-normalized determinant

$$\Delta \Gamma_{\text{PR}}^{(1)}[B] = \frac{1}{2} \text{STr}'_{\text{PR}} \log[H_{\text{PR}}[B] H_{\text{PR}}[0]^{-1}], \quad (137)$$

including vector, ghost, scalar, and fermion blocks [5]. Its electromagnetic charged-mode inventory supplies the exact Q_i^2 extraction for the photon kinetic term. The canonical neutral-response theorem developed here supplies the corresponding target-independent physical- Z closure at the declared finite tier.

The integrated PR-III statement is organized by the following stable labels:

- the determinant architecture, `eq:priii-reference-normalized-supertrace`;
- the fixed-kernel reduction before canonical normalization, `prop:priii-d3-structural-classification`;
- the exact charged-adjoint trace, `prop:priii-exact-charged-adjoint-trace`;
- the named canonical neutral-response closure H_Z , `eq:priii-neutral-response-hypothesis`;
- uniqueness of $k = 2$ in that declared response class, `thm:priii-conditional-k-two`;
- the relative-determinant Hessian, `lem:priii-determinant-hessian`;
- the excluded arbitrary- k second-jet family, `prop:priii-fixed-kernel-counterfamily`; and
- the ordinary W^3 -to- (Z, A) normalization distinction, `eq:priii-w3-to-physical-neutral-map`.

Together these results distinguish the exact group trace, the canonical physical-response closure, and its complementary determinant realization. The Step 06J generator implements the same theorem without a fitted numerical literal. It reads the locked charged weak-vector inventory, parses the eigencharges $(+1, -1)$ with exact **Fraction** arithmetic, and computes

$$k = (+1)^2 + (-1)^2 = \text{Tr}_{\text{ch}}(q^\dagger q) = 2. \quad (138)$$

Its generated provenance records that k is derived from the locked inventory and that the canonical neutral-response identity is a target-independent closure principle. A complementary equivalent microscopic representation is

$$\frac{1}{2} \text{STr}' \log(H/H_0) \implies \text{complete order-}\rho \text{ neutral Hessian} \implies \rho \mathcal{E}(K) \quad (139)$$

with unit prefactor, no other same-order term, and $\mathcal{E}(P_3) = P_Z$. It gives the same canonical response dynamically. With H1–H5 stated, the theorem is a complete canonical reduction of the declared fixed-kernel response domain; the three-gate family obtained by dropping H2 is outside the PR class.

In particular:

- $K = 2P_3$ is proved exactly from the frozen representation;
- every $k \neq 2$ has positive canonical-response defect $\rho^2(k-2)^2$;
- every tensor satisfying the full target-independent characterization is operator-equivalent to $2\rho P_Z$; and
- the larger three-gate class retains $D(k) = \rho k P_Z$, demonstrating why the minimal response principle must be part of the stated admissible class rather than inferred from diagnostic agreement.

The architecture-wide theorem includes the minimal canonical-response principle explicitly. The family in Eq. (130) is a scope witness outside that class, not an alternative PR solution. The accompanying public-source map retains the released manuscript anchors and current stable labels for full source-version reproducibility.

H Physical- Z Admissibility and Exclusion Theorem

This appendix verifies that the physical- Z normalization used by the canonical D_3 closure is fixed and that apparent second-jet or effective-field-theory alternatives acquire freedom only by leaving the PR admissible class. No diagnostic residual is used.

H.1 Exact Physical Neutral Connection

PR-II supplies

$$\mathcal{A}_\mu^{\text{neutral}} = gW_\mu^3 T_3 + g'B_\mu \frac{Y}{2}, \quad A = sW^3 + cB, \quad Z = cW^3 - sB, \quad (140)$$

where $e = gs = g'c$, $Q = T_3 + Y/2$, and $c^2 + s^2 = 1$.

Proposition H.1 (Physical- Z generator and normalized charged-vector trace). *Define $t_Z = T_3 - s^2 Q$, $g_Z = g/c$, and let $q_Z = \text{ad}_{g_Z t_Z}$ on $\text{span}\{T^+, T^-\}$. Then*

$$\mathcal{A}_\mu^{\text{neutral}} = eA_\mu Q + g_Z Z_\mu t_Z, \quad [g_Z t_Z, T^\pm] = \pm gc T^\pm. \quad (141)$$

Consequently the normalized physical charge operator $\hat{q}_Z = (gc)^{-1} q_Z$ has spectrum $(+1, -1)$ and

$$\text{Tr}_{\text{ch}}(\hat{q}_Z^\dagger \hat{q}_Z) = 2. \quad (142)$$

Proof. Inverting Eq. (140) gives $W^3 = sA + cZ$ and $B = cA - sZ$. The photon coefficient is $gsT_3 + g'cY/2 = eQ$, while the Z coefficient is $gcT_3 - g'sY/2 = (g/c)(T_3 - s^2Q)$. On the weak-vector adjoint $Y = 0$, so the ladder commutators have eigenvalues $\pm gc$. Division by gc gives the normalized spectrum and trace. $\square$

Thus the physical- Z map has exactly the unit-normalized charged-adjoint response assumed in Theorem G.5. It supplies $F(\hat{q}_Z) = 2$ without an adjustable mixing factor.

H.2 Second-jet Exclusion

For a normalized physical- Z background z , a reference-normalized determinant has second variation

$$\Gamma''(0) = \sum_i \eta_i \text{Tr}'_i (G_i W_i - G_i V_i G_i V_i), \quad V_i = H'_i(0), \quad W_i = H''_i(0). \quad (143)$$

Theorem H.2 (Independent Second Jets are Outside the PR Class). *Let one block be deformed by*

$$H_j^{(\lambda)}(z) = H_j^{(0)}(z) + \frac{\lambda\rho}{2\eta_j} z^2 H_j(0)^{1/2} X H_j(0)^{1/2}, \quad (144)$$

where X is positive, finite rank, commutes with the frozen charge action, and satisfies $\text{Tr}' X = 1$. The zero and first jets are unchanged, but

$$\Gamma''_\lambda(0) - \Gamma''_0(0) = \lambda\rho. \quad (145)$$

For every independent $\lambda \neq 0$, the deformation introduces a new response datum and therefore violates inheritance, minimality, and the no-new-coefficient rule of Proposition 2.2. It is not an alternative PR solution.

Proof. The deformation and its first derivative vanish at $z = 0$. Its second derivative changes only the $G_j W_j$ term in Eq. (143), by

$$\eta_j \text{Tr}' \left[H_j(0)^{-1} \frac{\lambda\rho}{\eta_j} H_j(0)^{1/2} X H_j(0)^{1/2} \right] = \lambda\rho \text{Tr}' X = \lambda\rho. \quad (146)$$

The free scalar λ is precisely an independent second-jet datum, which is forbidden by the completed minimal-response class. $\square$

H.3 Effective Operator Exclusion

Proposition H.3 (Independent neutral EFT operator is inadmissible). *For the PR-II $Y = 1$ doublet, the gauge-invariant operator*

$$\mathcal{O}_{HD} = |\Phi^\dagger D_\mu \Phi|^2, \quad \Delta\mathcal{L}_\lambda = \frac{2\lambda\rho}{v_{EW}^2} \mathcal{O}_{HD} \quad (147)$$

produces at $\Phi_0 = (0, v_{EW})^T/\sqrt{2}$

$$\Delta\mathcal{L}_\lambda|_{\Phi_0} = \frac{1}{2} \lambda\rho M_{Z,0}^2 Z_\mu Z^\mu, \quad M_{Z,0}^2 = \frac{1}{4} g_Z^2 v_{EW}^2. \quad (148)$$

It adds no charged-vector or photon mass, but its independent Wilson [9] coefficient λ and dimension-six operator violate the declared finite-tier operator inventory and no-new-parameter guideline. Hence it is a different theory, not a PR branch.

Proof. At the locked vacuum, $\Phi_0^\dagger D_\mu \Phi_0 = i v_{EW}^2 g_Z Z_\mu/4$. Taking the modulus square gives the displayed neutral mass term. The operator carries an independent Wilson coefficient and changes the scalar/Goldstone interactions, so it is excluded by the complete PR class definition. $\square$

Proposition H.4 (Physical-Z closure). *Every fixed-kernel neutral response satisfying the PR inheritance, photon nullity, charged-sector preservation, canonical unit normalization, and minimal quadratic-response guidelines is operator-equivalent to*

$$\boxed{D_3 = \rho \operatorname{Tr}_{\text{ch}}(\hat{q}_Z^\dagger \hat{q}_Z) P_Z = 2\rho P_Z.} \quad (149)$$

Every $k \neq 2$ has defect $\rho^2(k-2)^2$. Independent second jets and EFT operators have additional-data defects and lie outside $\mathfrak{A}_{\text{PR}}^{\text{can}}$. Therefore, neither construction supplies a second admissible coefficient, equation, or solution branch.

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